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Supplementary information

Structural pharmacology and therapeutic potential of 5-methoxytryptamines

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Structural pharmacology and therapeutic potential of 5-methoxytryptamines

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Supplementary Information includes:

Supplementary Data 1
Supplementary Table 1-3

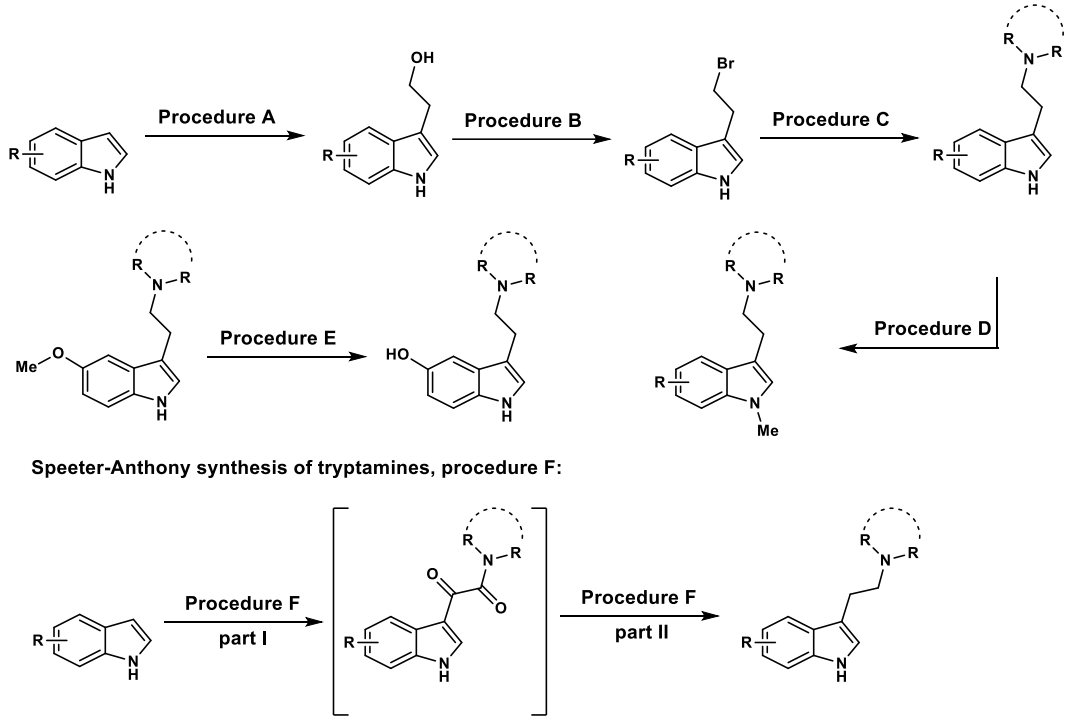
Supplementary Data 1 | Synthesis and Analysis of Tryptamine Compounds

Experimental Details

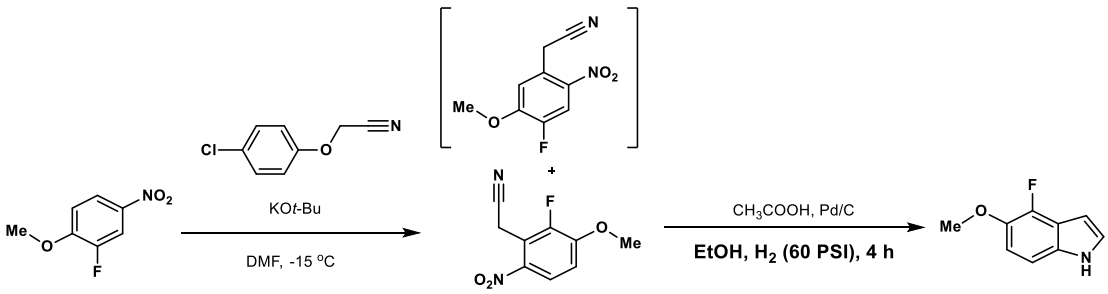
General Considerations.

Reagents and solvents (including anhydrous solvents) were obtained from commercial sources and used without further purification unless otherwise stated. Compounds not synthesized in-house were purchased from the following sources: 5-HT (Sigma), 5-MeO-DMT (Cayman Chemical), DMT (Cayman Chemical), Psilocin, 4-OH-MET (Cayman Chemical), LSD (Cayman Chemical), Mescaline (Cayman Chemical), Vilazodone (Sigma), Buspirone (Alfa Aesar), Bufotenine (Cayman Chemical), Aripiprazole (Sigma), Citalopram (TCI), FFN246 (Aobious), IDT307 (Sigma), ASP+ (Invitrogen), WAY 100,635 (Biovision), 8-OH-DPAT (MedChemExpress), SEP-363856 (MedChemExpress), N-(1-Napthyl)piperazine (Alfa Aesar), Tandospirone (Toronto Research Chemicals), Gepirone (MolPort). All reactions were performed in oven- or flame-dried glassware under an argon atmosphere unless otherwise stated and monitored by TLC using solvent mixtures appropriate to each reaction. All column chromatography was performed on silica gel (40-63 μm). Preparative TLC was performed on 20 x 20 cm plates with a 1 mm silica layer. For compounds containing a basic nitrogen, triethylamine (Et_3N) was used in the mobile phase to provide better resolution. Nuclear magnetic resonance spectra were recorded on Bruker 300, 400 or 500 MHz instruments as indicated. Chemical shifts are reported as δ values in ppm referenced to $CDCl_3$ (^{1}H NMR = 7.26 and ^{13}C NMR = 77.16). Multiplicity is indicated as follows: s (singlet); d (doublet); t (triplet); q (quartet); p (pentet); dd (doublet of doublets); ddd (doublet of doublet of doublets); dt (doublet of triplets); td (triplet of doublets); m (multiplet); br (broad). When certain carbon peaks overlap and thus represent two carbons they were indicated by (2C) designation. High-resolution mass spectra (HRMS) were acquired on a high-resolution Waters XEVO G2-XS QToF mass spectrometer equipped with a UPC2 SFC inlet, on-board fluidics and an ESI probe.

Scheme 1: General outline of tryptamine synthesis:

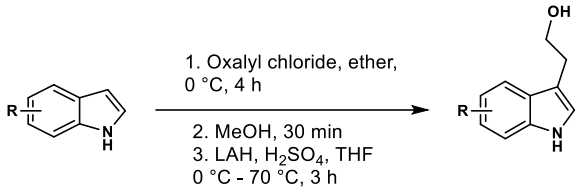


Scheme 2: Synthesis of 4-fluoro-5-methoxy-indole:



4-fluoro-5-methoxy-indole was synthesized according to the patent **WO03/064413** with minor modifications (highlighted). The regioisomeric intermediates (4-fluoro-5-methoxy-2-nitro-phenyl)-acetonitrile and (2-fluoro-3-methoxy-6-nitro-phenyl)-acetonitrile are separated by a sequence of crystallization of (4-fluoro-5-methoxy-2-nitro-phenyl)-acetonitrile from 2-propanol followed by crystallization of (2-fluoro-3-methoxy-6-nitro-phenyl)-acetonitrile (m/z $[M+H]^+$ = 166.1) from toluene using the mother liquid of the previous crystallization (as described in patent **WO 2007/096393)**.

Scheme 3: General procedure **A** for the synthesis of tryptophols [65]:

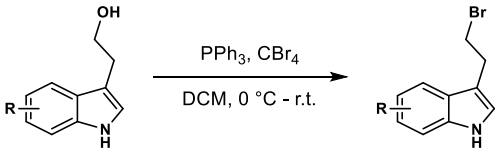


In a reaction flask under argon, 1 equiv. of indole was dissolved in dry diethyl ether (0.2 M), and oxalyl chloride (1.5 equiv.) was added dropwise to the reaction solution at 0 °C. The reaction was continued for 3-6 h at room temperature to form a suspension. The reaction mixture was further cooled to 0 °C, 20 equiv. of methanol (MeOH) was added dropwise to the reaction, and the mixture was stirred for 30 min. The resulting suspension was filtered and the collected solid washed with cold diethyl ether to finally obtain the glyoxyester as a fine powder. The powder was used for the next step without further purification.

Reduction Method I: A suspension of lithium aluminum hydride (LAH, 5 equiv.) in tetrahydrofuran (THF, 1 M) was cooled to 0 °C. To the light-gray suspension was added the glyoxyester as a thin suspension in THF (0.5 M) at a slow rate keeping the temperature at 0-5 °C. After the addition, the mixture was stirred at r.t. for 30 min and then heated to 70 °C for 3-4 h. After cooling to 0 °C, the mixture was quenched with 1 part of water, 1 part of 15% NaOH and 3 parts of water (w/w with respect to the LAH) and stirred until solids in resulting suspension were fine and loose. Suspension was filtered through a plug of Celite, plug was washed with ethyl acetate (EtOAc), and the combined washes were reduced in vacuum. The tryptophol product was purified by silica gel chromatography as described for each example.

Reduction method II: A suspension of LAH (5 equiv.) in THF (1 M) was cooled to 0 °C and a solution of H_2SO_4 (2.5 equiv.) in THF was added dropwise at a slow rate and the mixture was stirred for about 20 minutes. To the light-gray suspension was added the glyoxyester suspended in THF (0.5 M) at a slow rate keeping the temperature at 0-5 °C. After the addition the mixture was stirred at r.t. for 30 min and then heated to 70 °C for 4 h. Quenching and work-up were carried out identical to **Reduction method I**.

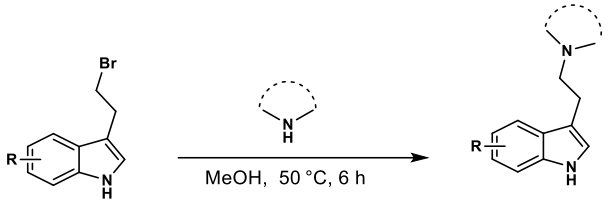
Scheme 4: General procedure **B** for the synthesis of 3-(2-bromoethyl)-indoles [65]:



The desired tryptophol and carbon tetrabromide (CBr_4) were dissolved in dry dichloromethane (DCM, 0.8 M), cooled to 0 °C, and triphenylphosphine (PPh_3) dissolved in dry DCM (2.5 M) was added dropwise over 10 min. The reaction mixture was stirred at r.t for 1-3 h (monitored by TLC).

The DCM was evaporated, and the crude mixture was purified by silica gel chromatography as described for each example.

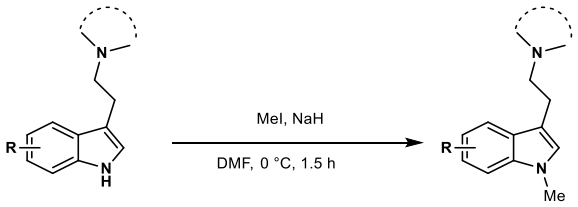
Scheme 5: General procedure **C** for tryptamines synthesis:



Method I: The desired 3-(2-bromoethyl)-1*H*-indole derivative was dissolved in MeOH and the appropriate alkyl amine (10 equiv.) was added in one portion. The resulting mixture was stirred at 50 °C for 4-6 h (monitored by TLC). At completion, the solvent was reduced in vacuum, and the crude mixture was purified by a silica gel column or preparative TLC plate as described for each example.

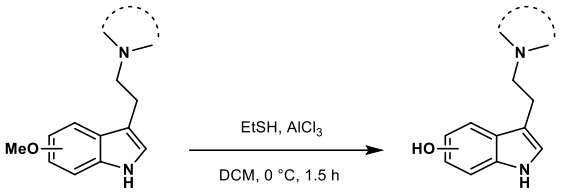
Method II [67]**:** The desired 3-(2-bromoethyl)-1*H*-indole derivative was dissolved in acetonitrile (MeCN, 0.2 M) and the appropriate alkyl amine (1.5 equiv.) was added together with 4 equiv. of $NaHCO_3$. The resulting mixture was stirred at 50-80 °C for 4-6 h (followed by TLC). At completion, the reaction mixture was filtered over Celite, the solvent was reduced in vacuum and the crude mixture was loaded on a silica gel column or preparative TLC plate as described for each example.

Scheme 6: General synthetic procedure **D** for *N*-methylindole tryptamines:



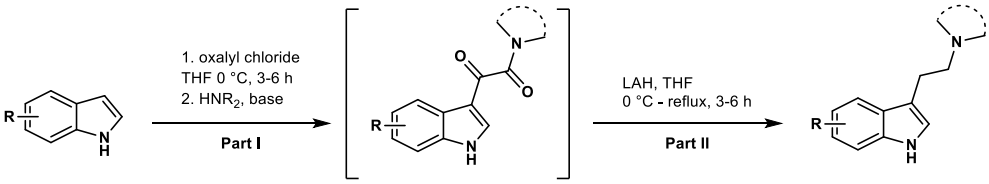
The desired tryptamine derivative was dissolved in *N*,*N*-Dimethylmethanamide (DMF, 0.1 M) and the mixture was cooled to 0 °C. Sodium hydride (NaH, 60% dispersion in mineral oil, 4 equiv.) was carefully added portion wise and the resulting mixture was stirred at r.t. for 1 h. Iodomethane (MeI, 1 equiv.) was added in one portion and the mixture was further stirred for 30 min. At completion, the reaction was quenched with water and extracted with diethyl ether (×3). The combined organic layers were dried over Na_2SO_4 and concentrated to provide the crude product. The crude mixture was purified by a silica gel column or preparative TLC plate as described for each example.

Scheme 7: General procedure **E** for hydroxy-indole tryptamines synthesis:



To a solution of the desired methoxy tryptamine (1 equiv.) in dry DCM (0.125 M) at 0 °C was added aluminum chloride ($AlCl_3$, 6 equiv.) followed by ethanethiol (EtSH, 18 equiv.), and the resulting mixture was allowed to warm to room temperature and stirred until TLC indicated the complete consumption of starting material (typically <1.5 h). The reaction was then quenched with saturated aqueous $NaHCO_3$ (100 mL per mmol of starting material) and extracted with DCM:iPrOH 9:1 (4x-6x, until no further extraction was observed by TLC). The combined organic layers were dried over Na_2SO_4 and concentrated to provide the crude product. The crude mixture was purified by a silica gel column or preparative TLC plate as described for each example.

Scheme 8: General procedure **F** – Speeter-Anthony synthesis of tryptamines:



Part I: In a reaction flask flushed with argon, solution of indole (1 equiv.) in dry diethyl ether (0.2 M) was treated dropwise with oxalyl chloride (1.5 equiv.) at 0 °C. The reaction was further continued for 3-6 h at room temperature to form a suspension. The solvent and excess oxalyl chloride were removed under reduced pressure and the residue dissolved in dry THF (1.0 M per indole), in some cases the suspension was filtered and the solid was used for the next steps without further purification. After cooling to 0 ˚C, a solution of the suitable N-substituted-amine* (3 or more equiv.) in dry THF (0.3 M) was added, and the resulting mixture was stirred at room temperature for 1.5 h. The solvent was removed under reduced pressure, the residue partitioned between EtOAc and water, and the aqueous phase extracted with EtOAc three times. The combined organic extracts were washed with brine, dried over Na_2SO_4 and concentrated under reduced pressure, to give a crude residue of the glyoxyamide.

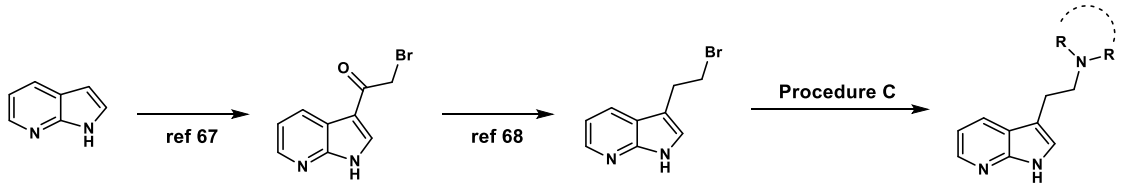
Part II: The crude glyoxyl amide material was suspended in dry THF (0.5 M) and slowly added to a suspension of LAH in dry THF (1 M) at 0 ˚C under argon**. The mixture was further refluxed for 3 h and after cooling to r.t. it was quenched with 1 part of water, 1 part of 15% NaOH and 3

parts of water (w/w with respect to the LAH). After the salts turned discolored and loose, solids were filtered and washed successively with EtOAc. The washes were combined and concentrated, and the crude residue was purified by a silica gel column or preparative TLC plate as described for each example.

* In some cases, triethylamine (TEA, 1.1 equiv.) was used instead of an excess of the desired amine.

** In some cases, the alane (AlH_3) was produced by the addition of H_2SO_4 to the LAH solution as described in general procedure **A**, **Reduction method II**.

Scheme 9: 7-azaindole tryptamine synthesis:



***N*-ethyl-2-(1*H*-indol-3-yl)-*N*-methylethan-1-amine (2)** Synthesized according to the general procedure C Method I. Using 3-(2-bromoethyl)-1*H*-indole (400 mg, 1.78 mmol, 1 equiv.) 9 mL of MeOH (0.2 M) and *N*-methyl-*N*-ethylamine (14.28 mmol, 1.2 mL, 8 equiv.). The product was isolated using column chromatography (3:1 Hexanes:EtOAc and 2% TEA) to yield 350 mg of the product, 97%. **^{1}H NMR** (400 MHz, MeOD) δ 7.53 (dt, *J* = 7.9, 1.0 Hz, 1H), 7.33 (dt, *J* = 8.2, 0.9 Hz, 1H), 7.13 – 7.05 (m, 1H), 7.00 (ddd, *J* = 8.0, 7.0, 1.1 Hz, 1H), 3.01 – 2.94 (m, 2H), 2.84 – 2.78 (m, 2H), 2.67 (q, *J* = 7.2 Hz, 2H), 2.43 (s, 3H), 1.16 (t, *J* = 7.2 Hz, 3H). **^{13}C NMR** (101 MHz, MeOD) δ 136.81, 127.19, 121.78, 120.95, 118.19, 117.69, 111.83, 110.88, 57.33, 50.78, 40.06, 21.98, 10.23. **HRMS** (ESI+): calcd. for $C_{13}H_{19}N_2$ $[M+H]^+$ = 203.1548, found $[M+H]^+$ = 203.1547.

***N*-(2-(1*H*-indol-3-yl)ethyl)-*N*-methylpropan-2-amine (3)** Synthesized according to the general procedure C Method I. Using 3-(2-bromoethyl)-1*H*-indole (45 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M) and *N*-methyl-*N*-isopropylamine (1.6 mmol, 0.2 mL, 8 equiv.). The product was isolated using column chromatography (3:1 Hexanes:EtOAc and 2% TEA) to yield 34 mg of the product, 78%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.03 (br s, 1H), 7.63 (d, *J* = 0.8 Hz, 1H), 7.36 (d, *J* = 0.9 Hz, 1H), 7.19 (ddd, *J* = 8.2, 7.0, 1.2 Hz, 1H), 7.12 (ddd, *J* = 8.0, 7.1, 1.1 Hz, 1H), 7.03 (d, *J* = 1.0 Hz, 1H), 3.03 – 2.87 (m, 3H), 2.79 – 2.69 (m, 2H), 2.37 (s, 3H), 1.06 (d, *J* = 6.5 Hz, 6H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.27, 127.60, 121.94, 121.41, 119.21, 118.90, 114.87, 111.10, 54.24, 53.57, 37.23, 24.20, 18.07. **HRMS** (ESI+): calcd. for $C_{14}H_{21}N_2$ $[M+H]^+$ = 217.1705, found $[M+H]^+$ = 217.1703.

***N*-(2-(1*H*-indol-3-yl)ethyl)-*N*-isopropylpropan-2-amine (4)** Synthesized according to the general procedure C Method I. 3-(2-bromoethyl)-1*H*-indole (60 mg, 0.27 mmol, 1 equiv.) was dissolved in 1.3 mL of MeOH (0.2 M), and diisopropylamine (2.68 mmol, 0.38 mL, 10 equiv.) was added. The product was isolated using column chromatography (4:1 Hexanes:EtOAc and 2% TEA) to yield 40 mg of the product, 61%. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 8.56 (s, 1H), 7.54 (dt, *J* = 7.8, 0.9 Hz, 1H), 7.43 (dt, *J* = 8.1, 0.9 Hz, 1H), 7.20 (ddd, *J* = 8.2, 7.0, 1.2 Hz, 1H), 7.16 – 7.09 (m, 2H), 3.72 (hept, *J* = 6.7 Hz, 2H), 3.53 – 3.42 (m, 2H), 3.24 – 3.13 (m, 2H), 1.53 (d, *J* = 6.5 Hz, 12H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.74, 127.39, 121.50, 120.83, 118.10, 117.88, 113.05, 110.87, 49.46, 27.52, 19.20. **HRMS** (ESI+): calcd. for $C_{16}H_{25}N_2$ $[M+H]^+$ = 245.2018, found $[M+H]^+$ = 245.2021.

3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (5) Synthesized according to the general procedure C Method I. Using 3-(2-bromoethyl)-1*H*-indole (100 mg, 0.45 mmol, 1 equiv.), 2 mL of MeOH (0.2 M) and pyrrolidine (4.03 mmol, 0.34 mL, 4.7 equiv.). The product was isolated using column chromatography (3:1 Hexanes:EtOAc and 2% TEA) to yield 13.6 mg of the product, 14%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.65 (s, 1H), 7.57 (d, *J* = 7.8 Hz, 1H), 7.41 (d, *J* = 8.1 Hz, 1H), 7.19 (t, *J* = 7.6 Hz, 1H), 7.14 – 7.06 (m, 2H), 3.47 – 2.95 (m, 8H), 2.13 – 1.99 (m, 4H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.20, 126.53, 122.68, 122.16, 119.52, 118.14, 111.51, 110.33, 55.92, 53.80, 23.27, 22.33. **HRMS** (ESI+): calcd. for $C_{14}H_{19}N_2$ $[M+H]^+$ = 215.1548, found $[M+H]^+$ = 215.1550.

3-(2-(piperidin-1-yl)ethyl)-1*H*-indole (6) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-1*H*-indole (500 mg, 2.23 mmol, 1 equiv.), 8.5 mL of MeOH (0.2 M) and piperidine (22.27 mmol, 2.2 mL, 5.2 equiv.). The product was isolated using column chromatography (EtOAc and 2% TEA) to yield 51.4 mg of the product, 10%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.44 (s, 1H), 7.62 (d, *J* = 7.9 Hz, 1H), 7.34 (d, *J* = 8.1 Hz, 1H), 7.18 (t, *J* = 7.6 Hz, 1H), 7.11 (t, *J* = 7.5 Hz, 1H), 6.99 (s, 1H), 3.07 – 2.98 (m, 2H), 2.79 – 2.68 (m, 2H), 2.68 – 2.47 (m, 4H), 1.77 – 1.63 (m, 4H), 1.62 – 1.41 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.36, 127.53, 121.94, 121.71, 119.21, 118.89, 114.17, 111.27, 60.02, 54.59, 25.80, 24.36, 22.69. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2$ $[M+H]^+$ = 229.1705, found $[M+H]^+$ = 229.1704.

2-(2-(1*H*-indol-3-yl)ethyl)-2-azabicyclo[2.2.2]oct-5-ene (7) Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-1*H*-indole (448 mg, 2.0 mmol, 1 equiv.), 10 mL of MeCN (0.2 M), 2-azabicyclo[2.2.2]oct-5-ene hydroiodide (474 mg, 2.0 mmol, 1 equiv.) and $NaHCO_3$ (672 mg, 8.0 mmol, 4 equiv.). The resulting mixture was heated to 80 °C for 48 h. The crude material was purified by silica gel column chromatography (gradient of 1:2, 1:1 to 3:1 EtOAc in hexanes and 2% TEA), followed by preparative TLC (3:1 EtOAc:hexanes and 2% TEA) to yield 377 mg of the product as beige

solid, 75%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.10 (s, 1H), 7.61 (d, *J* = 7.9 Hz, 1H), 7.34 (d, *J* = 8.1 Hz, 1H), 7.18 (t, *J* = 7.5 Hz, 1H), 7.11 (t, *J* = 7.4 Hz, 1H), 6.99 (s, 1H), 6.40 (t, *J* = 7.4 Hz, 1H), 6.33 – 6.24 (m, 1H), 3.54 – 3.47 (m, 1H), 3.12 (dd, *J* = 9.6, 2.1 Hz, 1H), 3.02 – 2.79 (m, 3H), 2.62 – 2.50 (m, 2H), 2.23 – 2.12 (m, 1H), 2.11 – 1.99 (m, 1H), 1.65 – 1.55 (m, 1H), 1.34 (tt, *J* = 12.2, 3.1 Hz, 1H), 1.31 – 1.21 (m, 1H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.31, 133.32, 132.03, 127.66, 121.99, 121.53, 119.25, 119.04, 114.83, 111.18, 59.43, 55.68, 52.84, 30.94, 26.74, 24.52, 22.22. **HRMS** (ESI+): calcd. for $C_{17}H_{21}N_2$ $[M+H]^+$ = 253.1705, found $[M+H]^+$ = 253.1707.

***N*-ethyl-2-(5-methoxy-1*H*-indol-3-yl)-*N*-methylethan-1-amine (9)** Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (700 mg, 2.75 mmol, 1 equiv.), 14 mL of MeOH (0.2 M) and *N*-methyl-*N*-ethylamine (27.55 mmol, 2.36 mL, 10 equiv.) was added. The product was isolated on column chromatography (9:1 EtOAc:MeOH and 2% TEA) to yield 256 mg of the product, 40%. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 8.59 (s, 1H), 7.21 (d, *J* = 8.8 Hz, 1H), 7.07 (d, *J* = 2.5 Hz, 1H), 6.96 (d, *J* = 2.4 Hz, 1H), 6.86 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.87 (s, 3H), 3.03 – 2.90 (m, 2H), 2.80 – 2.69 (m, 2H), 2.59 (q, *J* = 7.2 Hz, 2H), 2.40 (s, 3H), 1.15 (t, *J* = 7.2 Hz, 3H).**^{13}C NMR** (101 MHz, $CDCl_3$) δ 153.83, 131.64, 127.90, 122.57, 113.84, 111.97, 100.80, 57.77, 56.01, 51.35, 41.55, 23.24, 12.16. **HRMS** (ESI+): calcd. for $C_{14}H_{21}N_2O$ $[M+H]^+$ = 233.1654, found $[M+H]^+$ = 233.1664.

***N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylpropan-1-amine (10)** Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M), and N-methyl-*N*-propylamine (1.97 mmol, 0.2 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 20.0 mg of the product, 41%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.17 (br s, 1H), 7.19 (d, *J* = 8.7 Hz, 1H), 7.02 (d, *J* = 2.5 Hz, 1H), 6.95 (d, *J* = 2.3 Hz, 1H), 6.82 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.83 (s, 3H), 2.95 – 2.83 (m, 2H), 2.74 – 2.65 (m, 2H), 2.45 – 2.39 (m, 2H), 2.34 (s, 3H), 1.58 – 1.48 (m, 2H), 0.90 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.50, 131.16, 127.59, 122.05, 113.85, 111.68, 111.53, 100.43, 59.50, 57.91, 55.64, 41.96, 22.90, 20.14, 11.71. **HRMS** (ESI+): calcd. for $C_{15}H_{23}N_2O$ $[M+H]^+$ = 247.1810, found $[M+H]^+$ = 247.1812.

***N,N*-diethyl-2-(5-methoxy-1*H*-indol-3-yl)ethan-1-amine (11)** Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (100 mg, 0.39 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and diethylamine (3.94 mmol, 0.4 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 60.0 mg of the product, 58%. **^{1}H NMR** (500 MHz, MeOD) δ 7.22 (d, *J* = 8.7 Hz, 1H), 7.05 – 6.97 (m, 2H), 6.76 (dd, *J* = 8.7, 2.4 Hz, 1H), 3.81 (s, 3H), 2.94 – 2.85 (m, 2H), 2.84 – 2.79 (m, 2H), 2.72 (q, *J* = 7.2 Hz, 4H), 1.12 (t, *J* = 7.2 Hz, 6H. **^{13}C NMR** (126 MHz, MeOD) δ 153.56, 132.05, 127.55, 122.51, 111.98, 111.58, 111.11,

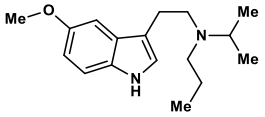
99.84, 54.91, 52.88, 46.36, 21.48, 9.79. **HRMS** (ESI+): calcd. for $C_{15}H_{23}N_2O$ $[M+H]^+$ = 247.1810, found $[M+H]^+$ = 247.1816.

***N*-ethyl-*N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)propan-1-amine (12)** Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M), and *N*-ethyl-*N*-propylamine (1.97 mmol, 0.19 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 37.0 mg of the product, 77%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.09 (br s, 1H), 7.28 – 7.17 (m, 1H), 7.06 (d, *J* = 2.5 Hz, 1H), 6.99 (d, *J* = 2.3 Hz, 1H), 6.86 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.87 (s, 3H), 2.96 – 2.88 (m, 2H), 2.85 – 2.77 (m, 2H), 2.69 (q, *J* = 7.2 Hz, 2H), 2.61 – 2.49 (m, 2H), 1.63 – 1.49 (m, 2H), 1.11 (t, *J* = 7.2 Hz, 3H), 0.93 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.81, 131.42, 127.92, 122.24, 114.37, 111.97, 111.78, 100.74, 55.89, 55.59, 54.02, 47.53, 22.88, 20.21, 12.03, 11.72. **HRMS** (ESI+): calcd. for $C_{16}H_{25}N_2O$ $[M+H]^+$ = 261.1967, found $[M+H]^+$ = 261.1966.

***N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-propylpropan-1-amine (13)** Synthesized according to general procedure C Method I. 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.) was dissolved in 1 mL of MeOH (0.2 M), and dipropylamine (1.97 mmol, 0.27 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 30.0 mg of the product, 56%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.00 (br s, 1H), 7.32 – 7.23 (m, 1H), 7.08 (d, *J* = 2.5 Hz, 1H), 7.02 (d, *J* = 2.4 Hz, 1H), 6.88 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.89 (s, 3H), 2.96 – 2.88 (m, 2H), 2.87 – 2.79 (m, 2H), 2.63 – 2.49 (m, 4H), 1.64 – 1.49 (m, 4H), 0.94 (t, *J* = 7.3 Hz, 6H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.88, 131.46, 127.99, 122.27, 114.50, 112.04, 111.81, 100.82, 56.25, 55.95, 54.67, 22.91, 20.27, 12.04. **HRMS** (ESI+): calcd. for $C_{17}H_{27}N_2O$ $[M+H]^+$ = 275.2123, found $[M+H]^+$ = 275.2119.

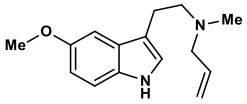
***N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylpropan-2-amine (14)** Synthesized according to general procedure F. Part I: 5-methoxy-1*H*-indole (192 mg, 1.31 mmol, 1 equiv.) was dissolved in 4.5 mL of THF (0.3 M), and oxalylchloride (1.96 mmol, 0.17 mL, 1.5 equiv.) was added. After filtration of the glyoxychloride intermediate, the material was resuspended in THF and *N*-methylisopropylamine (0.41 mL, 3.93 mmol, 3 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF and added dropwise to 1 M solution of LAH (249 mg, 6.56 mmol, 5 equiv.) in THF (6.5 mL) at 0 ˚C. The product was isolated on silica column (9:1 EtOAc:MeOH and 2% TEA) to yield 134 mg of the product, 41% over 2 steps. **^{1}H NMR** (500 MHz, MeOD) δ 7.21 (d, *J* = 8.8 Hz, 1H), 7.04 – 6.97 (m, 2H), 6.75 (dd, *J* = 8.8, 2.4 Hz, 1H), 3.79 (s, 3H), 2.93 (p, *J* = 6.5 Hz, 1H), 2.89 – 2.84 (m, 2H), 2.75 – 2.68 (m, 2H), 2.30 (s, 3H), 1.04 (d, *J* = 6.6 Hz, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 154.88, 133.36, 128.92, 123.91, 113.30, 112.96, 112.44, 101.24, 56.25, 55.13, 54.71, 37.08, 24.04, 17.82. **HRMS** (ESI+): calcd. for $C_{15}H_{23}N_2O$ $[M+H]^+$ = 247.1810, found $[M+H]^+$ = 247.1815.

***N*-isopropyl-*N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)propan-1-amine (15)** Synthesized



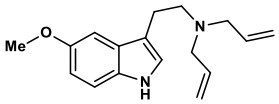
according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M), and *N*-isopropylpropan-1-amine (1.97 mmol, 0.27 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 29.0 mg of the product, 54%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.04 (br s, 1H), 7.24 (d, *J* = 8.8 Hz, 1H), 7.07 (d, *J* = 2.4 Hz, 1H), 7.00 (d, *J* = 2.4 Hz, 1H), 6.86 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.87 (s, 3H), 3.08 (p, *J* = 6.6 Hz, 1H), 2.91 – 2.84 (m, 2H), 2.79 – 2.70 (m, 2H), 2.52 – 2.46 (m, 2H), 1.59 – 1.50 (m, 2H), 1.06 (d, *J* = 6.6 Hz, 6H), 0.93 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.84, 131.44, 128.03, 122.31, 114.71, 111.99, 111.83, 100.85, 55.93, 52.60, 51.14, 50.89, 25.39, 22.29, 18.44, 12.08. **HRMS** (ESI+): calcd. for $C_{17}H_{27}N_2O$ $[M+H]^+$ = 275.2123, found $[M+H]^+$ = 275.2125.

***N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylprop-2-en-1-amine (16)** Synthesized



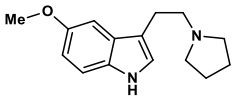
according to general procedure C Method I. 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.) was dissolved in 1 mL of MeOH (0.2 M), and *N*-methylprop-2-en-1-amine (1.97 mmol, 0.19 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 32.0 mg of the product, 67%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.17 (s, 1H), 7.22 (d, *J* = 8.8 Hz, 1H), 7.05 (d, *J* = 2.4 Hz, 1H), 6.98 (d, *J* = 2.4 Hz, 1H), 6.86 (dd, *J* = 8.7, 2.5 Hz, 1H), 6.01 – 5.87 (m, 1H), 5.28 – 5.13 (m, 2H), 3.87 (s, 3H), 3.18 – 3.10 (m, 2H), 2.99 – 2.90 (m, 2H), 2.79 – 2.70 (m, 2H), 2.38 (s, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.86, 135.53, 131.49, 127.89, 122.42, 117.83, 114.01, 112.07, 111.88, 100.75, 60.92, 57.66, 55.97, 42.07, 23.31. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2O$ $[M+H]^+$ = 245.1654, found $[M+H]^+$ = 245.1650.

***N*-allyl-*N*-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)prop-2-en-1-amine (17)** Synthesized according



to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M), and diallylamine (1.97 mmol, 0.27 mL, 10 equiv.). The product was isolated on preparative TLC (98:2 EtOAc:TEA) to yield 16 mg of the product, 30%. **^{1}H NMR** (500 MHz, MeOD) δ 7.20 (dd, *J* = 8.7, 0.6 Hz, 1H), 6.98 (d, *J* = 2.8 Hz, 2H), 6.74 (dd, *J* = 8.7, 2.5 Hz, 1H), 6.00 – 5.88 (m, 2H), 5.33 – 5.20 (m, 4H), 3.80 (s, 3H), 3.27 (dt, *J* = 6.8, 1.2 Hz, 4H), 2.91 – 2.87 (m, 2H), 2.83 – 2.77 (m, 2H). **^{13}C NMR** (126 MHz, MeOD) δ 154.87, 135.41, 133.38, 128.93, 123.89, 119.50, 113.39, 112.92, 112.52, 101.25, 57.64, 56.27, 54.73, 22.97. **HRMS** (ESI+): calcd. for $C_{17}H_{23}N_2O$ $[M+H]^+$ = 271.1810, found $[M+H]^+$ = 271.1808.

5-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (18) Synthesized according to general



procedure F with slight modifications. Part I: 5-methoxy-1*H*-indole (1.47 g, 9.99 mmol, 1 equiv.) was dissolved in 30 mL of THF (0.3 M), and

oxalylchloride (14.98 mmol, 1.28 mL, 1.5 equiv.) was added. After filtration of the glyoxychloride intermediate, the material was resuspended in THF and pyrrolidine (2.5 mL, 29.96 mmol, 3 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF (30 mL) and LAH was added portion wise (1.9 g, 49.95 mmol, 5 equiv., based on the indole starting material). The product was isolated using silica column chromatography (1:1 DCM:diethyl ether and 2% TEA) to yield 1.2 g of the product, 50% over 3 steps. **^{1}H NMR** (500 MHz, MeOD) δ 7.21 (d, *J* = 8.7 Hz, 1H), 7.04 – 6.96 (m, 2H), 6.75 (dd, *J* = 8.8, 2.5 Hz, 1H), 3.80 (s, 3H), 2.96 – 2.87 (m, 2H), 2.77 – 2.70 (m, 2H), 2.64 – 2.56 (m, 4H), 1.87 – 1.76 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 154.89, 133.39, 128.96, 123.78, 113.65, 112.89, 112.44, 101.30, 58.29, 56.29, 54.95, 25.66, 24.17. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2O$ $[M+H]^+$ = 245.1654, found $[M+H]^+$ = 245.1665.

5-methoxy-3-(2-(2-methylpyrrolidiN-1-yl)ethyl)-1*H*-indole (19) Synthesized according to general procedure C Method I. Using3-(2-bromoethyl)-5-methoxy-1*H*-indole (100 mg, 0.39 mmol, 1 equiv.), 1.5 mL of MeOH (0.2 M) and 2-methylpyrrolidine (3.9 mmol, 0.4 mL, 10 equiv.). The product was isolated on column chromatography (9:1 EtOAc:Hexanes and 2% TEA) to yield 88 mg of the product, 87%. **^{1}H NMR** (500 MHz, DMSO) δ 7.29 – 7.22 (m, 2H), 7.09 (d, *J* = 2.5 Hz, 1H), 6.75 (dd, *J* = 8.7, 2.5 Hz, 1H), 3.79 (s, 3H), 3.74 – 3.65 (m, 1H), 3.58 – 3.45 (m, 2H), 3.28 – 3.19 (m, 2H), 3.12 – 3.01 (m, 2H), 2.26 – 2.17 (m, 1H), 2.08 – 1.88 (m, 2H), 1.68 – 1.56 (m, 1H), 1.39 (d, *J* = 6.5 Hz, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 153.94, 132.05, 127.05, 123.52, 111.88, 111.57, 108.61, 99.68, 64.18, 55.03, 53.18, 30.81, 23.03, 21.74, 20.99, 16.08, 14.81. **HRMS** (ESI+): calcd. for $C_{16}H_{23}N_2O$ $[M+H]^+$ = 259.1810, found $[M+H]^+$ = 259.1814.

5-methoxy-3-(2-(piperidiN-1-yl)ethyl)-1*H*-indole (20) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (500 mg, 1.97 mmol, 1 equiv.), 8.5 mL of MeOH (0.2 M) and piperidine (22.27 mmol, 2.2 mL, 6.6 equiv.). The product was isolated on column chromatography (1:1 EtOAc:Hexanes and 2% TEA) to yield 330 mg of the product, 65%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.28 (s, 1H), 7.23 (d, *J* = 8.8 Hz, 1H), 7.06 (d, *J* = 2.4 Hz, 1H), 6.98 (d, *J* = 2.5 Hz, 1H), 6.84 (dd, *J* = 8.8, 2.4 Hz, 1H), 3.85 (s, 3H), 3.11 – 2.91 (m, 2H), 2.82 – 2.63 (m, 2H), 2.67 – 2.49 (m, 4H), 1.81 – 1.65 (m, 4H), 1.59 – 1.42 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.97, 131.53, 127.89, 122.55, 113.74, 112.19, 112.00, 100.79, 59.81, 56.09, 54.56, 25.67, 24.27, 22.67. **HRMS** (ESI+): calcd. for $C_{16}H_{23}N_2O$ $[M+H]^+$ = 259.1810, found $[M+H]^+$ = 259.1822.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-5-methoxy-1*H*-indole (21) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxy-1*H*-indole (109 mg, 0.43 mmol, 1 equiv.), 2 mL of MeOH and 3-pyrroline (30.43 mmol, 0.26 mL, 8 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 63 mg of the product, 61 %. **^{1}H NMR** (300 MHz, DMSO) δ

10.58 (s, 1H), 7.21 (d, *J* = 8.8 Hz, 1H), 7.10 (d, *J* = 2.3 Hz, 1H), 6.98 (d, *J* = 2.4 Hz, 1H), 6.70 (dd, *J* = 8.7, 2.4 Hz, 1H), 5.81 (s, 2H), 3.76 (s, 3H), 3.57 – 3.42 (m, 4H), 2.89 – 2.69 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 155.13, 133.57, 129.08, 128.28, 124.14, 113.24, 113.10, 112.70, 101.43, 60.80, 58.21, 56.47, 25.70. **HRMS** (ESI+): calcd. for $C_{15}H_{19}N_2O$ $[M+H]^+$ = 243.1497, found $[M+H]^+$ = 243.1495.

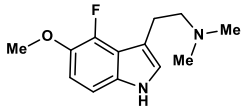
3-(2-(1*H*-imidazol-1-yl)ethyl)-5-methoxy-1*H*-indole (22) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)- 5-methoxy-1*H*-indole (81 mg, 0.32 mmol, 1 equiv.), 1.5 mL of MeOH, 1*H*-imidazole (0.32 mmol, 22 mg, 1 equiv.) and $NaHCO_3$ (87 mg, 1.27 mmol, 4 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 26 mg of the product, 34 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.47 (s, 1H), 7.23 (d, *J* = 8.7 Hz, 1H), 7.16 – 7.07 (m, 1H), 7.01 – 6.93 (m, 1H), 6.90 – 6.86 (m, 2H), 6.76 (dd, *J* = 8.7, 2.4 Hz, 1H), 4.32 (t, *J* = 6.7 Hz, 2H), 3.82 (s, 3H), 3.20 (t, *J* = 6.7 Hz, 2H). **^{13}C NMR** (126 MHz, MeOD) δ 153.73, 137.08, 131.87, 127.35, 127.31, 119.30, 111.57, 111.39, 110.48, 99.41, 54.87, 47.73, 27.07. **HRMS** (ESI+): calcd. for $C_{14}H_{16}N_3O$ $[M+H]^+$ = 242.1293, found $[M+H]^+$ = 242.1296.

3-(2-(3,6-dihydropyridin-1(2H)-yl)ethyl)-5-methoxy-1*H*-indole (23) Synthesized according to general procedure F. 5-methoxy-1*H*-indole (2.0 g, 12.1 mmol, 1 equiv.) was dissolved in 50 mL of dry diethyl ether (0.2 M), and oxalyl chloride (20.8 mmol, 1.75 mL, 1.5 equiv.) was added. The product was filtered and used directly in the next steps. The glyoxychloride (1 g, 4.2 mmol, 1 equiv.) was suspended in 1.0 M of THF and a mixture of TEA (0.87 mL, 6.3 mmol, 1.5 equiv.) and 1,2,3,6-tetrahydropyridine (0.46 mL, 5.0 mmol, 1.2 equiv.) in 20 mL THF (0.2 M) was added. The crude glyoxyamide was suspended in 8 mL of dry THF (0.5 M) and added dropwise to a cold suspension of LAH (5 equiv.) in dry THF (1.0 M). The crude material was purified on silica gel column chromatography (9:1 EtOAc:MeOH and 2% TEA) to yield 1.01 g of the product, 91 %. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.20 (s, 1H), 7.23 (dd, *J* = 8.8, 0.6 Hz, 1H), 7.07 (d, *J* = 2.5 Hz, 1H), 6.99 (d, *J* = 2.4 Hz, 1H), 6.85 (dd, *J* = 8.8, 2.4 Hz, 1H), 5.85 – 5.78 (m, 1H), 5.75 – 5.67 (m, 1H), 3.86 (s, 3H), 3.16 – 3.10 (m, 2H), 3.05 – 2.97 (m, 2H), 2.81 – 2.76 (m, 2H), 2.71 (t, *J* = 5.7 Hz, 2H), 2.32 – 2.22 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.88, 131.46, 127.87, 125.35, 125.22, 122.44, 114.00, 112.08, 111.90, 100.73, 59.22, 55.99, 52.72, 50.16, 26.18, 23.18. **HRMS** (ESI+): calcd. for $C_{16}H_{21}N_2O$ $[M+H]^+$ = 257.1654, found $[M+H]^+$ = 257.1661.

2-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-2-azabicyclo[2.2.2]oct-5-ene (24) Synthesized according to general procedure F with some modifications. 5-methoxy-1*H*-indole (200 mg, 1.36 mmol, 1 equiv.) was dissolved in 4 mL of dry diethyl ether (0.2 M), and oxalyl chloride (134 µL, 1.56 mmol, 1.15 equiv.) was added. The product was collected by filteration and used directly in the next steps. The glyoxychloride was suspended in MeCN (0.2 M, 2 mL) and a mixture of TEA (753 µL, 5.4 mmol,

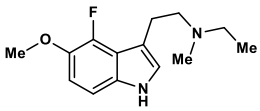
4 equiv. based on starting indole) and 2-azabicyclo[2.2.2]oct-5-ene hydroiodide (161 mg, 0.68 mmol, 0.5 equiv. based on starting indole) were added. The crude glyoxyamide was suspended in 7 mL of dry THF and added dropwise to a cold suspension of LAH (5 equiv.) in dry THF (1.0 M). The crude material was purified by repeated silica gel column chromatography (first column: gradient of 1:2, 1:1 to 2:1 AcOEt in hexanes and 2% TEA and second column: diethyl ether to diethyl ether and 2 % TEA) to yield 131 mg of the product as beige solid, 68 % over two steps. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.81 (s, 1H), 7.23 (d, *J* = 8.8 Hz, 1H), 7.05 (d, *J* = 2.4 Hz, 1H), 6.98 (d, *J* = 2.3 Hz, 1H), 6.84 (dd, *J* = 8.8, 2.5 Hz, 1H), 6.39 (t, *J* = 7.4 Hz, 1H), 6.31 – 6.25 (m, 1H), 3.86 (s, 3H), 3.54 – 3.46 (m, 1H), 3.11 (dd, *J* = 9.6, 2.1 Hz, 1H), 2.96 – 2.77 (m, 3H), 2.59 – 2.47 (m, 2H), 2.14 (dt, *J* = 9.6, 2.7 Hz, 1H), 2.10 – 1.97 (m, 1H), 1.65 – 1.54 (m, 1H), 1.38 – 1.29 (m, 1H), 1.29 – 1.20 (m, 1H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.9, 133.4, 131.9, 131.5, 128.0, 122.4, 114.4, 112.1, 111.9, 100.9, 59.2, 56.1, 55.6, 52.8, 30.9, 26.6, 24.4, 22.2. **HRMS** (ESI+): calcd. for $C_{18}H_{23}N_2O$ $[M+H]^+$ = 283.1810, found $[M+H]^+$ = 283.1822.

2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)-*N,N*-dimethylethan-1-amine (25) Synthesized



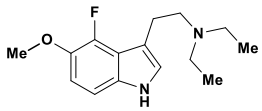
according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.) and 1.8 mL of 2 M dimethylamine solution in MeOH (3.67 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 60 mg of the product, 69%. **^{1}H NMR** (500 MHz, MeOD) δ 7.07 – 6.99 (m, 2H), 6.95 – 6.88 (m, 1H), 3.87 (s, 3H), 3.04 – 2.96 (m, 2H), 2.72 – 2.63 (m, 2H), 2.36 (s, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 147.58, 145.65, 139.28, 139.20, 134.92, 134.83, 123.55, 116.87, 116.74, 111.87, 110.49, 110.47, 106.37, 106.34, 60.88, 60.86, 58.18, 43.88, 23.79. **^{19}F NMR** (471 MHz, MeOD) δ -148.51. **HRMS** (ESI+): calcd. for $C_{13}H_{17}FN_2O$ $[M+H]^+$ = 237.1403, found $[M+H]^+$ = 237.1409.

***N*-ethyl-2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)-*N*-methylethan-1-amine (26)** Synthesized



according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and *N*-ethyl-*N*-methylamine (1.84 mmol, 0.16 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 23.0 mg of the product, 50%. **^{1}H NMR** (500 MHz, MeOD) δ 7.06 – 6.97 (m, 2H), 6.90 (t, *J* = 8.3 Hz, 1H), 3.86 (s, 3H), 3.03 – 2.92 (m, 2H), 2.75 – 2.65 (m, 2H), 2.55 (q, *J* = 7.7 Hz, 2H), 2.33 (s, 3H), 1.11 (t, *J* = 7.2 Hz, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 147.58, 145.65, 139.27, 139.19, 134.91, 134.83, 123.52, 116.91, 116.77, 111.82, 110.73, 110.71, 106.39, 106.36, 58.39, 58.37, 58.16, 50.67, 40.28, 23.18, 10.48. **^{19}F NMR** (471 MHz, MeOD) δ -148.39. **HRMS** (ESI+): calcd. for $C_{14}H_{20}FN_2O$ $[M+H]^+$ = 251.1560, found $[M+H]^+$ = 251.1568.

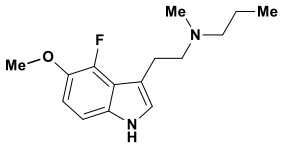
***N,N*-diethyl-2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethan-1-amine (27)** Synthesized according



to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.), 2 mL of MeOH and

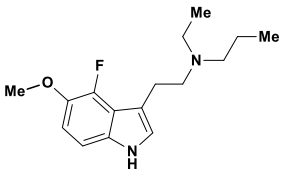
diethylamine (3.67 mmol, 0.3 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 92 mg of the product, 95%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.90 (s, 1H), 6.98 (d, *J* = 8.8 Hz, 1H), 6.94 – 6.85 (m, 2H), 3.05 – 2.94 (m, 2H), 2.86 – 2.78 (m, 2H), 2.70 (q, *J* = 7.2 Hz, 4H), 1.11 (t, *J* = 7.2 Hz, 6H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 147.80, 145.85, 139.75, 139.67, 134.39, 134.30, 123.41, 117.51, 117.37, 112.75, 112.73, 112.04, 106.46, 106.42, 59.03, 54.42, 54.41, 46.93, 23.64, 11.68. **^{19}F NMR** (471 MHz, MeOD) δ -148.45. **HRMS** (ESI+): calcd. for $C_{15}H_{22}FN_2O$ $[M+H]^+$ = 265.1716, found $[M+H]^+$ = 265.1728.

***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylpropan-1-amine (28)** Synthesized



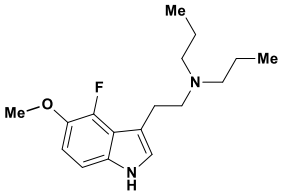
according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and *N*-methyl-*N*-propylamine (1.84 mmol, 0.18 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 40.0 mg of the product, 82%. **^{1}H NMR** (500 MHz, MeOD) δ 7.07 – 6.96 (m, 2H), 6.91 (t, *J* = 8.3 Hz, 1H), 3.87 (s, 3H), 3.03 – 2.93 (m, 2H), 2.76 – 2.64 (m, 2H), 2.47 – 2.38 (m, 2H), 2.33 (s, 3H), 1.65 – 1.48 (m, 2H), 0.92 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 148.98, 147.05, 140.65, 140.57, 136.28, 136.20, 124.87, 118.32, 118.19, 113.19, 112.23, 112.21, 107.76, 107.73, 60.56, 60.24, 59.54, 42.34, 24.53, 20.84, 12.21. **^{19}F NMR** (471 MHz, MeOD) δ -147.17. **HRMS** (ESI+): calcd. for $C_{15}H_{22}FN_2O$ $[M+H]^+$ = 265.1716, found $[M+H]^+$ = 265.1726.

***N*-ethyl-*N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)propan-1-amine (29)** Synthesized



according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and *N*-ethyl-*N*-propylamine (1.84 mmol, 0.23 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 44.0 mg of the product, 86%. **^{1}H NMR** (500 MHz, MeOD) δ 7.06 – 6.98 (m, 2H), 6.94 – 6.84 (m, 1H), 3.87 (s, 3H), 3.02 – 2.92 (m, 2H), 2.86 – 2.76 (m, 2H), 2.70 (q, *J* = 7.2 Hz, 2H), 2.60 – 2.50 (m, 2H), 1.62 – 1.50 (m, 2H), 1.12 (t, *J* = 7.2 Hz, 3H), 0.92 (d, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 148.97, 147.04, 140.63, 140.55, 136.29, 136.20, 124.87, 118.31, 118.17, 113.12, 112.34, 112.32, 107.76, 107.73, 59.52, 56.50, 56.05, 56.03, 48.40, 23.98, 20.47, 12.29, 11.46. **^{19}F NMR** (471 MHz, MeOD) δ -148.32. **HRMS** (ESI+): calcd. for $C_{16}H_{24}FN_2O$ $[M+H]^+$ = 279.1873, found $[M+H]^+$ = 279.1872.

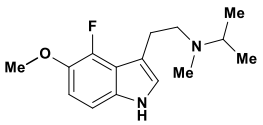
***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-propylpropan-1-amine (30)** Synthesized



according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and dipropylamine (1.84 mmol, 0.25 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 45.0 mg of the product, 84%. **^{1}H NMR** (500 MHz, MeOD) δ 7.08 – 6.95 (m, 2H), 6.89 (t, *J* = 8.3 Hz, 1H), 3.85 (s, 3H), 2.97 – 2.88 (m, 2H), 2.82 – 2.72 (m, 2H), 2.57 – 2.45 (m,

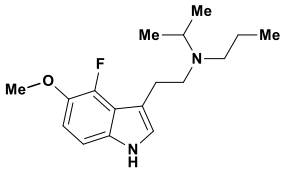
4H), 1.61 – 1.46 (m, 4H), 0.90 (t, *J* = 7.4 Hz, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 147.59, 145.66, 139.24, 139.17, 134.90, 134.82, 123.52, 116.93, 116.80, 111.73, 110.94, 110.92, 106.39, 106.35, 58.13, 55.74, 55.31, 55.29, 22.66, 19.16, 10.88. **^{19}F NMR** (471 MHz, MeOD) δ -148.35. **HRMS** (ESI+): calcd. for $C_{17}H_{26}FN_2O$ $[M+H]^+$ = 293.2029, found $[M+H]^+$ = 293.2039.

***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylpropan-2-amine (31)** Synthesized



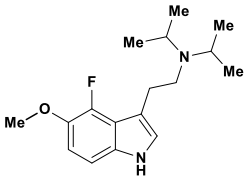
according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and *N*-methyl-*N*-isopropylamine (3.67 mmol, 0.4 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 97 mg of the product, 99%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.90 (s, 1H), 6.99 (d, *J* = 8.7 Hz, 1H), 6.95 – 6.86 (m, 2H), 3.92 (s, 3H), 3.05 – 2.89 (m, 3H), 2.82 – 2.72 (m, 2H), 2.38 (s, 3H), 1.06 (d, *J* = 6.6 Hz, 6H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 147.72, 145.77, 139.71, 139.63, 134.34, 134.26, 123.44, 117.46, 117.32, 112.62, 111.98, 106.43, 106.40, 58.96, 55.02, 55.00, 53.64, 37.27, 24.97, 18.01. **^{19}F NMR** (471 MHz, MeOD) δ -148.45. **HRMS** (ESI+): calcd. for $C_{15}H_{22}FN_2O$ $[M+H]^+$ = 265.1716, found $[M+H]^+$ = 265.1724.

***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-isopropylpropan-1-amine (32)**



Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and *N*-propyl-*N*-isopropylamine (1.84 mmol, 0.25 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 40.0 mg of the product, 74%. **^{1}H NMR** (500 MHz, MeOD) δ 7.04 – 6.97 (m, 2H), 6.89 (t, *J* = 8.3 Hz, 1H), 3.86 (s, 3H), 3.04 (p, *J* = 6.6 Hz, 1H), 2.95 – 2.88 (m, 2H), 2.76 – 2.69 (m, 2H), 2.53 – 2.46 (m, 2H), 1.58 – 1.48 (m, 2H), 1.05 (d, *J* = 6.6 Hz, 6H), 0.91 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 148.98, 147.05, 140.61, 140.53, 136.30, 136.21, 124.87, 118.33, 118.19, 113.10, 112.50, 112.48, 107.74, 107.71, 59.54, 53.71, 53.47, 53.46, 52.24, 26.51, 22.27, 18.48, 12.29. **^{19}F NMR** (471 MHz, MeOD) δ -148.30. **HRMS** (ESI+): calcd. for $C_{17}H_{26}FN_2O$ $[M+H]^+$ = 293.2029, found $[M+H]^+$ = 293.2039.

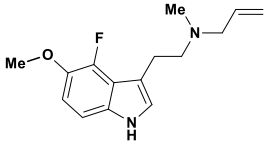
***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-isopropylpropan-2-amine (33)**



Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.), 4.0 mL of MeOH (0.2 M), and diisopropylamine (3.7mmol, 0.52 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 85 mg of the product, 79 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.06 – 6.96 (m, 2H), 6.89 (t, *J* = 8.3 Hz, 1H), 3.86 (s, 3H), 3.08 (hept, *J* = 6.5 Hz, 2H), 2.93 – 2.83 (m, 2H), 2.76 – 2.65 (m, 2H), 1.09 (d, *J* = 6.6 Hz, 12H). **^{13}C NMR** (126 MHz, MeOD) δ 148.97, 147.03, 140.57, 140.49, 136.29, 136.21, 124.75, 118.33, 118.20, 113.06, 112.69,

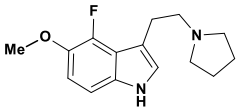
112.67, 107.70, 107.67, 59.56, 50.78, 49.59, 29.98, 20.53. **^{19}F NMR** (471 MHz, MeOD) δ -148.12. **HRMS** (ESI+): calcd. for $C_{17}H_{26}FN_2O$ $[M+H]^+$ = 293.2029, found $[M+H]^+$ = 293.2039.

***N*-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-*N*-methylprop-2-en-1-amine (34)**



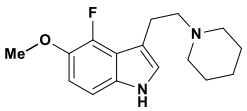
Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (50 mg, 0.18 mmol, 1 equiv.), 1 mL of MeOH and *N*-methyl-*N*-allylamine (1.84 mmol, 0.17 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 40.0 mg of the product, 74%. **^{1}H NMR** (500 MHz, MeOD) δ 7.05 – 6.97 (m, 2H), 6.90 (t, *J* = 8.3 Hz, 1H), 5.90 (ddt, *J* = 17.0, 10.2, 6.8 Hz, 1H), 5.28 – 5.15 (m, 2H), 3.86 (s, 3H), 3.16 – 3.09 (m, 2H), 3.00 – 2.95 (m, 2H), 2.75 – 2.65 (m, 2H), 2.33 (s, 3H). **^{13}C NMR** (126 MHz, MeOD) δ 147.59, 145.66, 139.26, 139.19, 134.90, 134.81, 134.08, 123.52, 117.91, 116.91, 116.77, 111.83, 110.67, 110.65, 106.38, 106.34, 60.18, 58.47, 58.46, 58.17, 40.64, 23.39. **^{19}F NMR** (471 MHz, MeOD) δ -148.35. **HRMS** (ESI+): calcd. for $C_{15}H_{20}FN_2O$ $[M+H]^+$ = 263.1560, found $[M+H]^+$ = 263.1568.

4-fluoro-5-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (35)



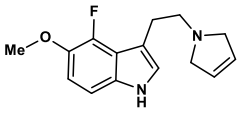
Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and pyrrolidine (3.67 mmol, 0.3 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 92 mg of the product, 95 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.08 – 6.98 (m, 1H), 6.96 – 6.86 (m, 2H), 3.87 (s, 3H), 3.08 – 2.97 (m, 2H), 2.83 – 2.74 (m, 2H), 2.69 – 2.59 (m, 4H), 1.88 – 1.78 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 148.95, 147.02, 140.66, 124.84, 113.17, 112.24, 107.75, 107.72, 59.52, 59.35, 54.96, 26.76, 24.15. **^{19}F NMR** (471 MHz, MeOD) δ -148.52. **HRMS** (ESI+): calcd. for $C_{15}H_{20}FN_2O$ $[M+H]^+$ = 263.1560, found $[M+H]^+$ = 263.1569.

4-fluoro-5-methoxy-3-(2-(piperidin-1-yl)ethyl)-1*H*-indole (36)



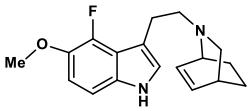
Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (100 mg, 0.37 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and piperidine (3.67 mmol, 0.42 mL, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 92 mg of the product, 95 %. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.92 (s, 1H), 6.93 (d, *J* = 8.7 Hz, 1H), 6.91 – 6.81 (m, 2H), 3.91 (s, 3H), 3.08 – 2.97 (m, 2H), 2.78 – 2.65 (m, 2H), 2.63 – 2.49 (m, 4H), 1.74 – 1.59 (m, 4H), 1.54 – 1.39 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 147.72, 145.78, 139.66, 139.58, 134.31, 134.23, 123.33, 117.38, 117.24, 112.63, 112.03, 106.33, 106.29, 60.92, 60.90, 59.01, 54.45, 25.83, 24.40, 23.57. **^{19}F NMR** (471 MHz, MeOD) δ -148.37. **HRMS** (ESI+): calcd. for $C_{16}H_{22}FN_2O$ $[M+H]^+$ = 277.1716, found $[M+H]^+$ = 277.1725.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-4-fluoro-5-methoxy-1*H*-indole (37) Synthesized



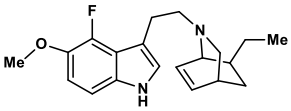
according to general procedure C Method II. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (55 mg, 0.2 mmol, 1 equiv.), 1.5 mL of MeCN, 3-pyrroline (0.22 mmol, 17.0 mg, 1.2 equiv.) and $NaHCO_3$ (68 mg, 0.8 mmol, 4 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 45 mg of the product, 86 %. **^{1}H NMR** (400 MHz, MeOD) δ 7.09 – 6.98 (m, 2H), 6.91 (t, *J* = 8.3 Hz, 1H), 5.85 – 5.77 (m, 2H), 3.86 (s, 3H), 3.61 – 3.52 (m, 4H), 3.10 – 2.89 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 147.59, 145.65, 139.28, 139.20, 134.90, 134.81, 126.84, 123.56, 116.91, 116.77, 111.86, 110.61, 106.38, 106.35, 59.21, 58.19, 57.88, 57.86, 25.44. **^{19}F NMR** (471 MHz, MeOD) δ -148.33. **HRMS** (ESI+): calcd. for $C_{15}H_{18}FN_2O$ $[M+H]^+$ = 261.1403, found $[M+H]^+$ = 261.1412.

2-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-2-azabicyclo[2.2.2]oct-5-ene (38)



Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (500 mg, 1.84 mmol, 1 equiv.), 15 mL of MeCN. 2-azabicyclo[2.2.2]oct-5-ene hydroiodide (2.2 mmol, 523 mg, 1.2 equiv.) and $NaHCO_3$ (772 mg, 9.19 mmol, 5 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 455 mg of the product, 82 %. **^{1}H NMR** (500 MHz, MeOD) δ 6.83 (d, *J* = 8.7 Hz, 1H), 6.78 (s, 1H), 6.70 (t, *J* = 8.3 Hz, 1H), 6.25 – 6.17 (m, 1H), 6.11 – 6.04 (m, 1H), 3.67 (s, 3H), 3.16 – 3.09 (m, 1H), 2.83 – 2.76 (m, 2H), 2.71 (ddd, *J* = 13.6, 11.6, 4.9 Hz, 1H), 2.58 (td, *J* = 11.6, 5.2 Hz, 1H), 2.38 – 2.28 (m, 2H), 1.98 (dt, *J* = 10.1, 2.7 Hz, 1H), 1.85 (ddd, *J* = 13.2, 8.3, 3.5 Hz, 1H), 1.36 (tt, *J* = 11.5, 2.9 Hz, 1H), 1.16 – 1.01 (m, 2H). ^{13}C NMR (126 MHz, MeOD) δ 148.97, 147.03, 140.62, 140.54, 136.24, 136.16, 134.66, 132.73, 124.81, 118.31, 118.17, 113.15, 112.35, 112.33, 107.73, 107.70, 61.31, 61.29, 59.55, 55.84, 53.57, 31.78, 26.24, 25.75, 22.77. **^{19}F NMR** (471 MHz, MeOD) δ -148.40. **HRMS** (ESI+): calcd. for $C_{18}H_{22}FN_2O$ $[M+H]^+$ = 301.1716, found $[M+H]^+$ = 301.1724.

7-ethyl-2-(2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethyl)-2-azabicyclo[2.2.2]oct-5-ene (39)



Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (500 mg, 1.84 mmol, 1 equiv.), 15 mL of MeCN (0.125 M), 2-azabicyclo[2.2.2]oct-5-ene hydroiodide (2.2 mmol, 584 mg, 1.2 equiv.) and $NaHCO_3$ (772 mg, 9.19 mmol, 5 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 455 mg of the product, 82 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.05 – 6.93 (m, 2H), 6.87 (t, *J* = 8.3 Hz, 1H), 6.35 – 6.24 (m, 2H), 3.85 (s, 3H), 3.04 (dd, *J* = 9.3, 2.3 Hz, 1H), 2.94 (ddd, *J* = 13.7, 10.2, 5.8 Hz, 1H), 2.83 (ddd, *J* = 13.8, 10.2, 4.8 Hz, 1H), 2.75 (td, *J* = 10.8, 5.8 Hz, 1H), 2.46 (td, *J* = 10.9, 4.8 Hz, 1H), 2.42 – 2.37 (m, 1H), 1.98 (dt, *J* = 9.3, 2.6 Hz, 1H), 1.62 – 1.43 (m, 3H), 1.34 – 1.22 (m, 1H), 0.93 – 0.84 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 149.15, 147.21, 140.51, 140.44, 136.24, 136.15, 133.98, 133.73, 124.71, 118.47, 118.33, 113.12, 113.10, 113.04, 107.58, 107.54, 61.29, 61.28, 59.64, 57.03,

56.80, 42.45, 32.84, 30.88, 28.09, 26.34, 12.88. **^{19}F NMR** (471 MHz, MeOD) δ -148.25. **HRMS** (ESI+): calcd. for $C_{20}H_{26}FN_2O$ $[M+H]^+$ = 329.2029, found $[M+H]^+$ = 329.2036.

3-(2-(dimethylamino)ethyl)-4-fluoro-1*H*-indol-5-ol (40) Synthesized according to general procedure E. Using 2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)-*N*,*N*-dimethylethan-1-amine (20 mg, 0.09 mmol, 1 equiv.), 0.7 mL of dry DCM (0.125 M), $AlCl_3$ (68 mg, 0.5 mmol, 6 equiv.) and ethanethiol (95 mg, 1.5 mmol, 18 equiv.). After workup, the product was isolated on preparative TLC (95:5 EtOAc:MeOH and 1% NH_3 sat. aq. sol.) to yield 10 mg of the product, 53%. **^{1}H NMR** (500 MHz, MeOD) δ 7.00 – 6.89 (m, 2H), 6.74 (t, *J* = 8.4 Hz, 1H), 3.04 – 2.96 (m, 2H), 2.73 – 2.65 (m, 2H), 2.37 (s, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 146.86, 144.97, 137.04, 136.95, 134.95, 134.87, 125.85, 124.47, 118.15, 114.46, 113.16, 111.47, 111.28, 111.26, 107.86, 107.83, 62.23, 62.21, 45.20, 25.13, 14.46. **^{19}F NMR** (471 MHz, MeOD) δ -152.48. **HRMS** (ESI+): calcd. for $C_{12}H_{16}FN_2O$ $[M+H]^+$ = 223.1247, found $[M+H]^+$ = 223.1251.

3-(2-(diethylamino)ethyl)-4-fluoro-1*H*-indol-5-ol (41) Synthesized according to general procedure E. Using *N*,*N*-diethyl-2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethan-1-amine (20 mg, 0.08 mmol, 1 equiv.), 0.7 mL of dry DCM (0.125 M), $AlCl_3$ (68 mg, 0.5 mmol, 6 equiv.) and ethanethiol (95 mg, 1.5 mmol, 18 equiv.). After workup, the product was isolated on preparative TLC (95:5 EtOAc:MeOH with 1% NH_3 sat. aq. sol.) to yield 6 mg of the product, 32%. **^{1}H NMR** (500 MHz, MeOD) δ 7.02 – 6.90 (m, 2H), 6.75 (t, *J* = 8.4 Hz, 1H), 3.04 – 2.97 (m, 2H), 2.95 – 2.88 (m, 2H), 2.81 (q, *J* = 7.2 Hz, 4H), 1.17 (t, *J* = 7.2 Hz, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 146.83, 144.94, 137.09, 137.00, 134.88, 124.62, 117.95, 114.52, 111.00, 107.94, 107.91, 55.21, 47.98, 30.79, 23.70, 10.93. **^{19}F NMR** (471 MHz, MeOD) δ -152.52. **HRMS** (ESI+): calcd. for $C_{14}H_{20}FN_2O$ $[M+H]^+$ = 251.1560, found $[M+H]^+$ = 251.1560.

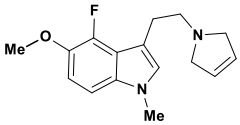
4-fluoro-3-(2-(pyrrolidiN-1-yl)ethyl)-1*H*-indol-5-ol (42) Synthesized according to general procedure E. Using 4-fluoro-5-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (20 mg, 0.08 mmol, 1 equiv.), 0.7 mL of dry DCM (0.125 M), $AlCl_3$ (68 mg, 0.5 mmol, 6 equiv.) and ethanethiol (95 mg, 1.5 mmol, 18 equiv.). After workup, the product was isolated on preparative TLC (95:5 EtOAc:MeOH with 1% NH_3 sat. aq. sol.) to yield 12 mg of the product, 63%. **^{1}H NMR** (500 MHz, MeOD) δ 7.00 – 6.89 (m, 2H), 6.74 (t, *J* = 8.4 Hz, 1H), 3.07 – 2.97 (m, 2H), 2.87 – 2.80 (m, 2H), 2.72 – 2.64 (m, 4H), 1.89 – 1.81 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 146.87, 144.98, 137.05, 136.96, 134.93, 134.84, 124.38, 118.17, 118.03, 114.44, 111.62, 111.60, 107.83, 107.80, 59.36, 59.34, 55.01, 26.68, 24.16. **^{19}F NMR** (471 MHz, MeOD) δ -152.75. **HRMS** (ESI+): calcd. for $C_{14}H_{18}FN_2O$ $[M+H]^+$ = 249.1403, found $[M+H]^+$ = 249.1404.

5-methoxy-1-methyl-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (43) Synthesized according to general procedure D. Using 5-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (80 mg, 0.33 mmol, 1 equiv.), 2.5 mL of DMF, NaH (60% dispersion in mineral oil, 1.31 mmol, 52 mg, 4 equiv.) and MeI (0.33 mmol, 46 mg, 1 equiv.). The product was isolated on preparative TLC (EtOAc and 2% TEA) to yield 70 mg of the product, 83 %. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.17 (d, *J* = 8.8 Hz, 1H), 7.06 (d, *J* = 2.4 Hz, 1H), 6.91 – 6.85 (m, 2H), 3.86 (s, 3H), 3.71 (s, 3H), 2.77 – 2.65 (m, 4H), 1.93 – 1.83 (m, 4H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 153.71, 132.40, 128.02, 127.09, 111.74, 109.99, 100.87, 57.29, 56.07, 54.25, 32.78, 29.72, 23.48. **HRMS** (ESI+): calcd. for $C_{16}H_{23}N_2O$ $[M+H]^+$ = 259.1810, found $[M+H]^+$ = 259.1811.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-5-methoxy-1-methyl-1*H*-indole (44), 3-(2-(1*H*-pyrrol-1-yl)ethyl)-5-methoxy-1-methyl-1*H*-indole (45) Synthesized according to general procedure D. Using 3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-5-methoxy-1*H*-indole (20 mg, 0.082 mmol, 1 equiv.), 0.8 mL of DMF, NaH (60% dispersion in mineral oil, 0.33 mmol, 13 mg, 4 equiv.) and MeI (0.082 mmol, 12 mg, 1 equiv.). The products were isolated on preparative TLC (EtOAc and 2% TEA to yield 3 mg of **44**, 14 % and 8 mg of **45**, 38%. **44: ^{1}H NMR** (500 MHz, MeOD) δ 7.18 (dd, *J* = 8.8, 3.1 Hz, 1H), 7.02 (d, *J* = 2.4 Hz, 1H), 6.92 (d, *J* = 4.3 Hz, 1H), 6.80 (dd, *J* = 8.8, 2.4 Hz, 1H), 5.82 (d, *J* = 2.9 Hz, 2H), 3.81 (d, *J* = 1.1 Hz, 3H), 3.67 (s, 3H), 3.58 – 3.49 (m, 4H), 2.98 – 2.83 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 155.06, 134.06, 129.45, 128.31, 128.28, 112.58, 112.49, 110.93, 101.63, 60.59, 58.16, 56.30, 32.74, 25.57. **HRMS** (ESI+): calcd. for $C_{16}H_{21}N_2O$ $[M+H]^+$ = 257.1654, found $[M+H]^+$ = 257.1651. **45: ^{1}H NMR** (500 MHz, MeOD)δ 7.17 (dd, *J* = 8.9, 0.6 Hz, 1H), 6.84 (d, *J* = 2.4 Hz, 1H), 6.79 (dd, *J* = 8.8, 2.4 Hz, 1H), 6.73 (s, 1H), 6.61 (t, *J* = 2.1 Hz, 2H), 5.99 (t, *J* = 2.1 Hz, 2H), 4.11 (t, *J* = 7.0 Hz, 2H), 3.79 (s, 3H), 3.66 (s, 3H), 3.10 (t, *J* = 0.8 Hz, 2H). **^{13}C NMR** (126 MHz, MeOD) δ 155.10, 133.91, 129.45, 128.85, 121.50, 112.57, 112.07, 110.80, 108.54, 101.44, 56.30, 51.48, 32.73, 28.91. **HRMS** (ESI+): calcd. for $C_{16}H_{19}N_2O$ $[M+H]^+$ = 255.1497, found $[M+H]^+$ = 255.1500.

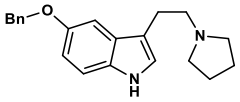
4-fluoro-5-methoxy-1-methyl-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (46) Synthesized according to general procedure D. Using 4-fluoro-5-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (20 mg, 0.076 mmol, 1 equiv.), 0.7 mL of DMF, NaH (60% dispersion in mineral oil, 0.3 mmol, 12 mg, 4 equiv.) and MeI (0.076 mmol, 11 mg, 1 equiv.). The product was isolated on preparative TLC (EtOAc and 2% TEA) to yield 5 mg of the product, 24 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.07 – 6.92 (m, 3H), 3.88 (s, 3H), 3.70 (s, 3H), 3.05 – 2.96 (m, 2H), 2.89 – 2.80 (m, 2H), 2.78 – 2.67 (m, 4H), 1.95 – 1.78 (m, 4H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 149.98, 148.02, 141.90, 141.82, 137.29, 137.20, 129.83, 119.88, 119.74, 114.11, 113.73, 113.71, 106.53, 106.50, 61.29, 60.48, 56.43, 35.04, 28.23, 25.71. **^{19}F NMR** (471 MHz, $CDCl_3$) δ -145.86. **HRMS** (ESI+): calcd. for $C_{16}H_{22}FN_2O$ $[M+H]^+$ = 277.1716, found $[M+H]^+$ = 277.1723.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-4-fluoro-5-methoxy-1-methyl-1*H*-indole (47)



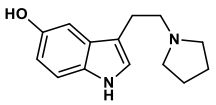
Synthesized according to general procedure D. Using 3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-4-fluoro-5-methoxy-1*H*-indole (20 mg, 0.076 mmol, 1 equiv.), 0.7 mL of DMF, NaH (60% dispersion in mineral oil, 0.3 mmol, 12 mg, 4 equiv.) and MeI (0.076 mmol, 11 mg, 1 equiv.). The product was isolated on preparative TLC (EtOAc and 2% TEA) to yield 6 mg of the product, 28 %. **^{1}H NMR** (500 MHz, MeOD) δ 7.07 – 6.93 (m, 3H), 5.91 – 5.78 (m, 2H), 3.87 (s, 3H), 3.70 (s, 3H), 3.66 – 3.60 (m, 4H), 3.05 – 2.96 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 147.41, 139.58, 135.26, 128.13, 126.60, 117.02, 111.78, 109.34, 104.48, 104.45, 59.25, 58.03, 57.69, 57.67, 31.48, 24.98. ^{19}F NMR (471 MHz, $CDCl_3$) δ -145.94. **HRMS** (ESI+): calcd. for $C_{16}H_{20}FN_2O$ $[M+H]^+$ = 275.1560, found $[M+H]^+$ = 275.1567.

5-(benzyloxy)-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (48) Synthesized according to general



procedure F. 5-phenoxy-1*H*-indole (21.9 g, 104.6 mmol, 1 equiv.) was dissolved in 500 mL of dry diethyl ether (0.2 M), and oxalyl chloride (146.5 mmol, 12.57 mL, 1.4 equiv.) was added. The product was filtered and used directly in the next steps. The glyoxychloride (0.5 g, 1.67 mmol, 1 equiv.) was suspended in 1.0 M of THF and a mixture of TEA (0.35 mL, 2.5 mmol, 1.5 equiv.) and pyrrolidine (0.17 mL, 2.0 mmol, 1.2 equiv.) in 8 mL THF (0.2 M) were added. The crude glyoxyamide was suspended in 4 mL of dry THF (0.5 M) and added dropwise to a cold suspension of LAH (5 equiv.) in dry THF (1.0 M). The crude material was purified using silica gel column chromatography (9:1 EtOAc:MeOH and 2% TEA) to yield 405 mg of the product, 76 % over 2 steps. **^{1}H NMR** (500 MHz, MeOD) δ 7.40 – 7.35 (m, 2H), 7.30 – 7.25 (m, 2H), 7.24 – 7.19 (m, 2H), 7.07 (d, *J* = 2.4 Hz, 1H), 6.95 (s, 1H), 6.83 (dd, *J* = 8.8, 2.4 Hz, 1H), 5.00 (s, 2H), 2.88 – 2.80 (m, 2H), 2.68 – 2.59 (m, 2H), 2.52 – 2.42 (m, 4H), 1.72 (p, *J* = 3.2 Hz, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 153.83, 139.35, 133.51, 129.34, 128.91, 128.57, 128.55, 123.88, 113.63, 113.21, 112.93, 103.33, 71.96, 58.14, 54.81, 25.61, 24.10. **HRMS** (ESI+): calcd. for $C_{21}H_{25}N_2O$ $[M+H]^+$ = 321.1967, found $[M+H]^+$ = 321.1964.

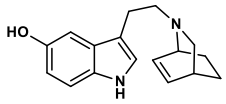
3-(2-(pyrrolidine-1-yl)ethyl)-1*H*-indol-5-ol (50) 5-methoxy-3-(2-(pyrrolidine-1-yl)ethyl)-1*H*-



indole (25 mg, 0.10 mmol, 1 equiv.) was dissolved in 1 mL of DCM (0.100 M). BBr_3 (0.20 mmol, 205 µL, 2 equiv.) was added to the solution in a -70 ˚C dry ice and acetone bath. This yellow solution was stirred for 16 hours before quenching with distilled water (100 mL per mmol of starting material). The organic layer was extracted using 9:1 DCM:iPrOH 4×-6×, until no further visible by TLC), dried using Na_2SO_4, and concentrated to provide the crude product. This mixture was loaded on a preparative TLC plate and the product was isolated (9:1 DCM:iPrOH with 2% TEA) to yield 7.8 mg of the product, 33%. **^{1}H NMR** (500 MHz, MeOD) δ 7.19 (d, *J* = 8.6 Hz, 1H), 7.03 (s, 1H), 6.95 (d, *J* = 2.3 Hz, 1H), 6.69 (dd, *J* = 8.6, 2.3 Hz, 1H), 3.00 – 2.91 (m, 2H), 2.91 – 2.83 (m, 2H), 2.81 – 2.72 (m, 4H), 1.96 –

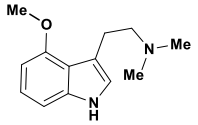
1.87 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 151.99, 133.98, 130.14, 124.81, 113.55, 113.22, 104.27, 59.09, 55.91, 26.36, 25.04. **HRMS** (ESI+): calcd. for $C_{14}H_{19}N_2O$ $[M+H]^+$ = 231.1497, found $[M+H]^+$ = 231.1493.

3-(2-(2-azabicyclo[2.2.2]oct-5-eN-2-yl)ethyl)-1*H*-indol-5-ol (51) Synthesized according to



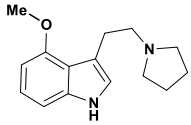
general procedure E. Using 2-(2-(5-methoxy-1*H*-indol-3-yl)ethyl)-2-azabicyclo[2.2.2]oct-5-ene (67 mg, 0.24 mmol, 1 equiv.), 1.8 mL of dry DCM (0.125 M), $AlCl_3$ (190 mg, 1.42 mmol, 6 equiv.) and ethanethiol (307 µL, 4.26 mmol, 18 equiv.). After workup, the crude material was purified by silica gel column chromatography (gradient of 10 to 50% MeOH in acetone) followed by preparative TLC (20% MeOH in DCM) to yield 22 mg of the product, 35 %. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.70 (s, 1H), 8.08 (s, 1H), 7.12 (d, *J* = 8.6 Hz, 1H), 6.91 – 6.81 (m, 2H), 6.77 (dd, *J* = 8.6, 2.3 Hz, 1H), 6.35 (t, *J* = 7.4 Hz, 1H), 6.19 (dd, *J* = 8.1, 5.4 Hz, 1H), 3.59 – 3.50 (m, 1H), 3.11 (dd, *J* = 10.1, 2.1 Hz, 1H), 3.01 – 2.87 (m, 2H), 2.87 – 2.78 (m, 1H), 2.63 – 2.53 (m, 1H), 2.53 – 2.47 (m, 1H), 2.18 – 2.06 (m, 2H), 1.63 – 1.53 (m, 1H), 1.31 – 1.14 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 150.7, 133.8, 131.5, 131.3, 128.2, 122.7, 113.2, 113.0, 112.0, 103.7, 59.1, 54.8, 53.0, 30.6, 25.6, 23.8, 21.8. **HRMS** (ESI+): calcd. for $C_{17}H_{21}FN_2O$ $[M+H]^+$ = 269.1654, found $[M+H]^+$ = 269.1650.

2-(4-methoxy-1*H*-indol-3-yl)-N,N-dimethylethaN-1-amine (54) Synthesized according to



general procedure C Method I. Using 3-(2-bromoethyl)-4-methoxy-1*H*-indole (100 mg, 0.39 mmol, 1 equiv.) and 2 mL of 2 M dimethylamine solution in MeOH (3.9 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 44.0 mg of the product, 51%. **^{1}H NMR** (500 MHz, MeOD) δ 6.99 (t, *J* = 7.9 Hz, 1H), 6.95 – 6.89 (m, 2H), 6.45 (d, *J* = 7.6 Hz, 1H), 3.89 (s, 3H), 3.13 – 3.02 (m, 2H), 2.87 – 2.77 (m, 2H), 2.47 (s, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 155.66, 139.98, 123.31, 122.32, 118.31, 112.90, 105.88, 99.75, 62.42, 55.32, 44.94, 25.17. **HRMS** (ESI+): calcd. for $C_{13}H_{19}N_2O$ $[M+H]^+$ = 219.1497, found $[M+H]^+$ = 219.1489.

4-methoxy-3-(2-(pyrrolidiN-1-yl)ethyl)-1*H*-indole (55) Synthesized according to general



procedure C Method I. Using 3-(2-bromoethyl)- 4-methoxy-1*H*-indole (100 mg, 0.39 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and pyrrolidine (0.32 mL, 3.9 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 45.0 mg of the product, 47%. **^{1}H NMR** (500 MHz, MeOD) δ 6.97 (t, *J* = 7.9 Hz, 1H), 6.92 (d, *J* = 8.2 Hz, 1H), 6.88 (s, 1H), 6.42 (d, *J* = 7.6 Hz, 1H), 3.86 (s, 3H), 3.11 – 3.03 (m, 2H), 2.84 – 2.75 (m, 2H), 2.71 – 2.61 (m, 4H), 1.87 – 1.76 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 155.74, 139.89, 123.18, 121.99, 118.41, 113.92, 105.80, 99.67, 59.80, 55.28, 54.96, 27.06, 24.15. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2O$ $[M+H]^+$ = 245.1654, found $[M+H]^+$ = 245.1663.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-4-methoxy-1*H*-indole (56) Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-4-methoxy-1*H*-indole (100 mg, 0.39 mmol, 1 equiv.), 4 mL of MeCN (0.1 M), 3-pyrroline (41.0 mg, 0.59 mmol, 1.5 equiv.) and $NaHCO_3$ (135 mg, 1.57 mmol, 4 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 27.0 mg of the product, 28%. **^{1}H NMR** (500 MHz, MeOD) δ 6.97 (t, *J* = 7.9 Hz, 1H), 6.91 (dd, *J* = 8.2, 0.9 Hz, 1H), 6.87 (s, 1H), 6.42 (d, *J* = 7.6 Hz, 1H), 5.80 (s, 2H), 3.86 (s, 3H), 3.57 – 3.50 (m, 4H), 3.05 – 2.99 (m, 2H), 2.94 – 2.88 (m, 2H). **^{13}C NMR** (126 MHz, MeOD) δ 155.74, 139.89, 128.25, 123.14, 122.01, 118.44, 113.95, 105.79, 99.65, 60.54, 59.86, 55.28, 27.26. **HRMS** (ESI+): calcd. for $C_{15}H_{19}N_2O$ $[M+H]^+$ = 243.1497, found $[M+H]^+$ = 243.1508.

2-(4-fluoro-1*H*-indol-3-yl)-*N,N*-dimethylethan-1-amine (57) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-1*H*-indole (200 mg, 0.39 mmol, 1 equiv.), and 4 mL of 2 M dimethylamine solution in MeOH). The product was isolated on preparative TLC (9:1 EtOAc:MeOH with 2% TEA) to yield 110 mg of the product, 54%. **^{1}H NMR** (500 MHz, MeOD) δ 7.16 (d, *J* = 8.2 Hz, 1H), 7.09 (s, 1H), 7.06 – 7.00 (m, 1H), 6.68 (dd, *J* = 11.5, 7.8 Hz, 1H), 3.13 – 3.04 (m, 2H), 2.92 – 2.83 (m, 2H), 2.50 (s, 6H). **^{13}C** NMR (126 MHz, MeOD) δ 159.31, 157.37, 141.23, 141.14, 124.09, 122.94, 122.88, 116.84, 111.05, 108.78, 108.75, 104.58, 104.42, 61.66, 61.64, 44.84, 24.67. **^{19}F NMR** (471 MHz, MeOD) δ -126.15. **HRMS** (ESI+): calcd. for $C_{12}H_{16}FN_2$ $[M+H]^+$ = 207.1297, found $[M+H]^+$ = 207.1303.

4-fluoro-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (58) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-4-fluoro-1*H*-indole (200 mg, 0.39 mmol, 1 equiv.), 4 mL of MeOH (0.2 M), and pyrrolidine (0.68 mL, 3.9 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 83 mg of the product, 37%. **^{1}H NMR** (500 MHz, MeOD) δ 7.12 (d, *J* = 8.1 Hz, 1H), 7.05 – 6.93 (m, 2H), 6.68 – 6.59 (m, 1H), 3.08 – 2.97 (m, 2H), 2.84 – 2.74 (m, 2H), 2.68 – 2.60 (m, 4H), 1.87 – 1.75 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 159.22, 157.29, 141.02, 140.92, 123.48, 122.59, 122.53, 116.99, 116.83, 112.21, 112.18, 108.46, 108.43, 104.28, 104.12, 59.12, 59.10, 54.79, 26.60, 23.98. **^{19}F NMR** (471 MHz, MeOD) δ -125.82. **HRMS** (ESI+): calcd. for $C_{14}H_{18}FN_2$ $[M+H]^+$ = 233.1454, found $[M+H]^+$ = 233.1459.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-4-fluoro-1*H*-indole (59) Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-4-fluoro-1*H*-indole (25 mg, 0.1 mmol, 1 equiv.), 0.5 mL of MeCN (0.1 M), 3-pyrroline (8.0 mg, 0.11 mmol, 1.1 equiv.) and $NaHCO_3$ (35 mg, 0.4 mmol, 4 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 9.0 mg of the product, 32%. **^{1}H NMR** (500 MHz, MeOD) δ 7.13 (d, *J* = 8.1 Hz, 1H), 7.04 (s, 1H), 7.03 – 6.97 (m, 1H), 6.64 (dd, *J* = 11.5, 7.7 Hz, 1H), 5.83 (d, *J* = 1.1 Hz, 2H), 3.60 (d, *J* = 1.2 Hz, 4H), 3.07 – 2.94 (m, 4H) **^{13}C**

NMR (126 MHz, MeOD) δ 159.55, 157.62, 141.38, 128.31, 123.98, 122.97, 122.91, 112.16, 108.84, 108.81, 104.65, 104.49, 60.79, 59.35, 59.33, 26.94. **^{19}F NMR** (471 MHz, MeOD) δ -125.97. **HRMS** (ESI+): calcd. for $C_{14}H_{16}FN_2$ $[M+H]^+$ = 231.1297, found $[M+H]^+$ = 231.1295.

2-(6-methoxy-1*H*-indol-3-yl)-*N,N*-dimethylethan-1-amine (60) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-6-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.) and 1 mL of 2 M dimethylamine solution in MeOH (1.97 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 37.0 mg of the product, 86%. **^{1}H NMR** (500 MHz, MeOD) δ 7.46 (d, *J* = 8.6 Hz, 1H), 7.06 (d, *J* = 0.9 Hz, 1H), 6.92 (d, *J* = 2.3 Hz, 1H), 6.72 (dd, *J* = 8.6, 2.3 Hz, 1H), 3.79 (s, 3H), 3.28 – 3.21 (m, 2H), 3.14 – 3.07 (m, 2H), 2.79 (s, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 157.83, 138.94, 122.84, 122.57, 119.69, 110.35, 110.26, 95.64, 59.51, 55.97, 43.86, 22.38. **HRMS** (ESI+): calcd. for $C_{13}H_{19}N_2O$ $[M+H]^+$ = 219.1497, found $[M+H]^+$ = 219.1491.

6-methoxy-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (61) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-6-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 1 mL of MeOH (0.2 M), and pyrrolidine (0.16 mL, 3.9 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 37.0 mg of the product, 77%. **^{1}H NMR** (500 MHz, MeOD) δ 7.38 (d, *J* = 8.6 Hz, 1H), 6.90 (d, *J* = 0.9 Hz, 1H), 6.86 (d, *J* = 2.2 Hz, 1H), 6.68 (dd, *J* = 8.6, 2.3 Hz, 1H), 3.77 (s, 3H), 2.95 – 2.88 (m, 2H), 2.79 – 2.73 (m, 2H), 2.65 – 2.57 (m, 4H), 1.85 – 1.75 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 156.19, 137.43, 121.79, 120.38, 118.42, 112.32, 108.42, 94.16, 56.98, 54.56, 53.57, 24.24, 22.78. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2O$ $[M+H]^+$ = 245.1654, found $[M+H]^+$ = 245.1656.

3-(2-(2,5-dihydro-1*H*-pyrrol-1-yl)ethyl)-6-methoxy-1*H*-indole (62) Synthesized according to general procedure C Method II. Using 3-(2-bromoethyl)-6-methoxy-1*H*-indole (50 mg, 0.2 mmol, 1 equiv.), 2 mL of MeCN (0.1 M), 3-pyrroline (15.0 mg, 0.22 mmol, 1.1 equiv.) and $NaHCO_3$ (70 mg, 0.8 mmol, 4 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 34 mg of the product, 71%. **^{1}H NMR** (500 MHz, MeOD) δ 7.38 (d, *J* = 8.6 Hz, 1H), 6.91 (s, 1H), 6.86 (d, *J* = 2.3 Hz, 1H), 6.68 (dd, *J* = 8.6, 2.3 Hz, 1H), 5.79 (s, 2H), 3.78 (s, 3H), 3.55 – 3.45 (m, 4H), 2.95 – 2.83 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 157.55, 138.78, 128.22, 123.18, 121.80, 119.80, 113.56, 109.78, 95.53, 60.55, 58.24, 55.93, 25.77. **HRMS** (ESI+): calcd. for $C_{15}H_{19}N_2O$ $[M+H]^+$ = 243.1497, found $[M+H]^+$ = 243.1491.

2-(5*H*-[1,3]dioxolo[4,5-*f*]indol-7-yl)-*N,N*-dimethylethan-1-amine (63) Synthesized according to general procedure C Method I. 7-(2-bromoethyl)-5*H*-[1,3]dioxolo[4,5-f]indole (100 mg, 0.37 mmol, 1 equiv.) was dissolved in 1.86 mL of 2 M

dimethylamine solution in MeOH (3.7 mmol, 10 equiv.). The product was isolated on preparative TLC (98:2 EtOAc:TEA) to yield 53.0 mg of the product, 61%. **^{1}H NMR** (400 MHz, MeOD) δ 6.94 – 6.84 (m, 2H), 6.79 (d, *J* = 0.6 Hz, 1H), 5.83 (s, 2H), 2.88 – 2.76 (m, 2H), 2.64 – 2.54 (m, 2H), 2.31 (s, 6H). **^{13}C NMR** (101 MHz, MeOD) δ 145.89, 143.65, 132.98, 122.58, 121.71, 113.87, 101.49, 97.73, 92.96, 61.34, 45.35, 24.31. **HRMS** (ESI+): calcd. for $C_{13}H_{17}N_2O_2$ $[M+H]^+$ = 233.1290, found $[M+H]^+$ = 233.1290.

7-(2-(pyrrolidine-1-yl)ethyl)-5*H*-[1,3]dioxolo[4,5-*f*]indole (64) Synthesized according to general procedure C Method I. Using 7-(2-bromoethyl)-5*H*-[1,3]dioxolo[4,5-f]indole (120 mg, 0.45 mmol, 1 equiv.), 2 mL of MeOH (0.2 M), and pyrrolidine (0.37 mL, 4.5 mmol, 10 equiv.).The product was isolated on preparative TLC (98:2 EtOAc:TEA) to yield 102.0 mg of the product, 88%. **^{1}H NMR** (400 MHz, MeOD) δ 6.92 – 6.85 (m, 2H), 6.79 (d, *J* = 0.6 Hz, 1H), 5.83 (s, 2H), 2.90 – 2.81 (m, 2H), 2.74 – 2.68 (m, 2H), 2.61 – 2.52 (m, 4H), 1.85 – 1.75 (m, 4H). **^{13}C NMR** (101 MHz, MeOD) δ 145.87, 143.62, 132.94, 122.59, 121.66, 114.16, 101.48, 97.80, 92.96, 58.35, 54.94, 25.72, 24.17. **HRMS** (ESI+): calcd. for $C_{15}H_{19}N_2O_2$ $[M+H]^+$ = 259.1447, found $[M+H]^+$ = 259.1450.

7-(2-(2-azabicyclo[2.2.2]oct-5-en-2-yl)ethyl)-5H-[1,3]dioxolo[4,5-f]indole (65) Synthesized according to general procedure C Method II. Using 7-(2-bromoethyl)-5*H*-[1,3]dioxolo[4,5-f]indole (1.0 g, 3.73 mmol, 1 equiv.), 30 mL of MeCN (0.125 M), 2-azabicyclo[2.2.2]oct-5-ene hydroiodide (1.06 g, 4.48 mmol, 1.2 equiv.) and $NaHCO_3$ (1.57 g, 18.65 mmol, 5 equiv.). The product was isolated using column chromatography (1:1 EtOAc:hexane with 2% TEA) to yield 844 mg of the product, 76%. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 8.28 (br s, 1H), 6.95 (s, 1H), 6.83 (t, *J* = 1.4 Hz, 1H), 6.79 – 6.69 (m, 1H), 6.42 – 6.36 (m, 1H), 6.32 – 6.22 (m, 1H), 5.91 (s, 2H), 3.55 – 3.44 (m, 1H), 3.10 (dd, *J* = 9.7, 2.1 Hz, 1H), 2.94 – 2.71 (m, 3H), 2.58 – 2.46 (m, 2H), 2.20 – 2.11 (m, 1H), 2.09 – 1.99 (m, 1H), 1.65 – 1.51 (m, 1H), 1.40 – 1.20 (m, 2H). **^{13}C NMR** (101 MHz, $CDCl_3$) δ 144.85, 142.63, 133.33, 132.02, 131.18, 121.54, 120.40, 114.79, 100.58, 97.60, 92.12, 59.30, 55.58, 52.78, 30.89, 26.57, 24.53, 22.20. **HRMS** (ESI+): calcd. for $C_{18}H_{21}N_2O$ $[M+H]^+$ = 297.1603, found $[M+H]^+$ = 297.1609.

7-(2-(7-ethyl-2-azabicyclo[2.2.2]oct-5-en-2-yl)ethyl)-5*H*-[1,3]dioxolo[4,5-*f*]indole (66) Synthesized according to general procedure C Method II. Using 7-(2-bromoethyl)-5*H*-[1,3]dioxolo[4,5-*f*]indole (1.0 g, 3.73 mmol, 1 equiv.), 30 mL of MeCN (0.125 M), 7-ethyl-2-azabicyclo[2.2.2]oct-5-ene hydroiodide (1.19 g, 4.48 mmol, 1.2 equiv.) and $NaHCO_3$ (1.57 g, 18.65 mmol, 5 equiv.). The product was isolated on column chromatography (1:1 EtOAc:hexane and 2% TEA) to yield 1.0 g of the product, 83%. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.84 (br s, 1H), 6.97 (s, 1H), 6.89 (d, *J* = 2.3 Hz, 1H), 6.79 (s, 1H), 6.40 – 6.24 (m, 2H), 5.92 (s, 2H), 3.31 – 3.25 (m, 1H), 3.13 (dd, *J* = 9.2, 2.3 Hz, 1H), 2.86 – 2.68 (m, 3H), 2.57 – 2.40 (m, 2H), 1.97 (dt, *J* = 9.2, 2.6 Hz, 1H), 1.69 – 1.45 (m, 3H), 1.38 – 1.29 (m, 1H), 0.96 (dd, *J* = 4.9, 2.2 Hz, 1H), 0.92 (t, *J* = 7.4 Hz, 3H). **^{13}C NMR** (101

MHz, $CDCl_3$) δ 144.78, 142.57, 132.87, 132.72, 130.97, 121.65, 120.26, 115.28, 100.51, 97.60, 91.99, 59.00, 56.25, 55.96, 41.18, 31.63, 29.77, 27.21, 24.49, 12.55. **HRMS** (ESI+): calcd. for $C_{20}H_{25}N_2O$ $[M+H]^+$ = 325.1916, found $[M+H]^+$ = 325.1927.

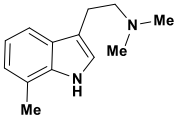
***N,N*-dimethyl-2-(5-(methylthio)-1*H*-indol-3-yl)ethan-1-amine (67)** Synthesized according to general procedure F. Part I: 5-(Methylthio)-1*H*-indole was used to prepare a stock of its corresponding glyoxychloride intermediate in the same manner described in the general procedure. The glyoxychloride (150 mg, 0.59 mmol, 1 equiv.) was resuspended in THF (2 mL, 0.3 M) and dimethylamine (1.8 mL of 2 M sol. in THF, 3.55 mmol, 6 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF and added dropwise to 1 M solution of LAH (131 mg, 3.45 mmol, 8 equiv.) and H_2SO_4 (963 uL, 1.72 mmol, 4 equiv.) in THF at 0 °C. The product was isolated on silica column (9:1 EtOAc:MeOH and 2% TEA) to yield 30 mg of the product, 30% over 2 steps. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.47 (br s, 1H), 7.64 – 7.57 (m, 1H), 7.28 – 7.23 (m, 1H), 7.20 (dd, *J* = 8.4, 1.7 Hz, 1H), 6.98 (d, *J* = 2.3 Hz, 1H), 2.99 – 2.92 (m, 2H), 2.73 – 2.66 (m, 2H), 2.52 (s, 3H), 2.39 (s, 6H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 137.04, 130.22, 129.28, 126.11, 124.28, 121.86, 115.98, 113.67, 62.13, 47.38, 25.49, 21.03. **HRMS** (ESI+): calcd. for $C_{13}H_{18}N_2S$ $[M+H]^+$ = 235.1269, found $[M+H]^+$ = 235.1267.

***N*-ethyl-*N*-methyl-2-(5-(methylthio)-1*H*-indol-3-yl)ethan-1-amine (68)** Synthesized according to general procedure F. Part I: 5-(Methylthio)-1*H*-indole was used to prepare a stock of its corresponding glyoxychloride intermediate in the same manner described in the general procedure. The glyoxychloride (150 mg, 0.59 mmol, 1 equiv.) was resuspended in THF (2 mL, 0.3 M) and N-ethyl-*N*-methylamine (0.3 mL, 3.55 mmol, 6 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF and added dropwise to 1 M solution of LAH (131 mg, 3.45 mmol, 8 equiv.) and H_2SO_4 (963 µL, 1.72 mmol, 4 equiv.) in THF at 0 °C. The product was isolated on silica column (9:1 EtOAc:MeOH and 2% TEA) to yield 126 mg of the product, 77% over 2 steps. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.41 (br s, 1H), 7.61 (d, *J* = 1.7 Hz, 1H), 7.27 – 7.18 (m, 2H), 6.98 (d, *J* = 2.2 Hz, 1H), 3.01 – 2.86 (m, 2H), 2.78 – 2.67 (m, 2H), 2.57 (q, *J* = 7.2 Hz, 2H), 2.52 (s, 3H), 2.37 (s, 3H), 1.12 (t, *J* = 7.2 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 137.28, 130.45, 129.37, 126.25, 124.58, 122.03, 116.17, 113.93, 59.95, 53.54, 43.70, 25.22, 21.24, 14.36. **HRMS** (ESI+): calcd. for $C_{14}H_{20}N_2S$ $[M+H]^+$ = 249.1425, found $[M+H]^+$ = 249.1420.

5-(methylthio)-3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-indole (69) Synthesized according to general procedure F. Part I: 5-(Methylthio)-1*H*-indole was used to prepare a stock of its corresponding glyoxychloride intermediate in the same manner described in the general procedure. The glyoxychloride (150 mg, 0.59 mmol, 1 equiv.) was resuspended in THF (2 mL, 0.3 M) and pyrrolidine (0.3 mL, 3.55 mmol, 6 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF and added dropwise to 1 M solution of LAH (131 mg, 3.45 mmol, 8 equiv.) and H_2SO_4 (963 µL, 1.72 mmol, 4 equiv.) in THF

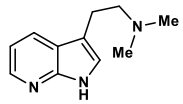
at 0 ˚C. The product was isolated on silica column (9:1 EtOAc:MeOH and 2% TEA) to yield 128 mg of the product, 75% over 2 steps. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.49 (br s, 1H), 7.62 (d, *J* = 1.7 Hz, 1H), 7.24 – 7.15 (m, 2H), 6.98 – 6.95 (m, 1H), 3.01 – 2.94 (m, 2H), 2.85 – 2.79 (m, 2H), 2.69 – 2.63 (m, 4H), 2.51 (s, 3H), 1.88 – 1.80 (m, 4H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 136.79, 129.88, 128.80, 125.69, 124.12, 121.56, 115.69, 113.39, 58.79, 55.87, 26.57, 25.17, 20.73. **HRMS** (ESI+): calcd. for $C_{15}H_{21}N_2S$ $[M+H]^+$ = 261.1425, found $[M+H]^+$ = 261.1435.

***N,N*-dimethyl-2-(7-methyl-1*H*-indol-3-yl)ethan-1-amine (70)** Synthesized according to



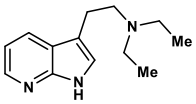
general procedure F. Part I: 7-methyl-1*H*-indole was used to prepare a stock of its corresponding glyoxychloride intermediate in the same manner described in the general procedure. The glyoxychloride (80 mg, 0.36 mmol, 1 equive.) was resuspended in THF (1.2 mL, 0.3 M) and dimethylamine (1.8 mL of 2 M sol. in THF, 3.55 mmol, 6 equiv.) was added. Part II: The glyoxyamide intermediate was suspended in THF and added dropwise to 1 M solution of LAH (108 mg, 2.88 mmol, 8 equiv.) and H_2SO_4 (787 uL, 1.44 mmol, 4 equiv.) in THF at 0 °C. The product was isolated on silica column (9:1 EtOAc:MeOH and 2% TEA) to yield 7 mg of the product, 10% over 2 steps. **^{1}H NMR** (400 MHz, MeOD) δ 7.42 – 7.31 (m, 1H), 7.05 (d, *J* = 0.9 Hz, 1H), 6.95 – 6.84 (m, 2H), 2.99 – 2.89 (m, 2H), 2.77 – 2.67 (m, 2H), 2.46 (s, 3H), 2.38 (s, 6H). **^{13}C NMR** (101 MHz, MeOD) δ 126.86, 121.58, 121.44, 120.33, 119.07, 118.46, 115.47, 112.28, 59.90, 43.83, 22.80, 15.45. **HRMS** (ESI+): calcd. for $C_{13}H_{19}N_2$ $[M+H]^+$ = 203.1548, found $[M+H]^+$ = 203.1542.

***N,N*-dimethyl-2-(*1H*-pyrrolo[2,3-b]pyridiN-3-yl)ethaN-1-amine (72)** Synthesized according to



general procedure C, Method I. Using 3-(2-bromoethyl)-1*H*-pyrrolo[2,3-b]pyridine (prepared according to the description in Scheme 9[67,68], 30 mg, 0.13 mmol, 1 equiv.) and 0.67 mL of 2 M dimethylamine solution in MeOH (1.33 mmol, 10 equiv.). The product was isolated on preparative TLC (9:1 EtOAc:MeOH and 2% TEA) to yield 14 mg of the product, 56%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.15 (dd, *J* = 4.8, 1.5 Hz, 1H), 8.00 (dd, *J* = 7.9, 1.5 Hz, 1H), 7.21 (d, *J* = 1.0 Hz, 1H), 7.08 (dd, *J* = 7.9, 4.8 Hz, 1H), 3.00 – 2.86 (m, 2H), 2.71 – 2.59 (m, 2H), 2.33 (s, 6H).**^{13}C NMR** (126 MHz, $CDCl_3$) δ 148.06, 141.64, 127.22, 122.85, 120.41, 114.71, 111.68, 59.69, 43.97, 22.76. **HRMS** (ESI+): calcd. for $C_{11}H_{16}N_3$ $[M+H]^+$ = 190.1344, found $[M+H]^+$ = 190.1350.

***N,N*-diethyl-2-(*1H*-pyrrolo[2,3-b]pyridin-3-yl)ethan-1-amine (73)** Synthesized according to



general procedure C, Method I. Using 3-(2-bromoethyl)-1*H*-pyrrolo[2,3-b]pyridine (prepared according to the description in Scheme 9,[67,68] 30 mg, 0.13 mmol, 1 equiv.), 0.67 mL of MeOH (0.2 M) and diethylamine (1.33 mmol, 0.14 mL, 10 equiv.). The product was isolated on column chromatography (1:1 EtOAc:Hexanes and 2% TEA) as the modifier to yield 14 mg of the product, 50%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.16 (dd, *J* = 4.8, 1.5 Hz, 1H), 7.99 (dd, *J* = 7.9, 1.6 Hz, 1H), 7.22 (s, 1H), 7.08 (dd, *J* = 7.8, 4.8 Hz, 1H), 2.97 – 2.85 (m, 2H), 2.85 – 2.76 (m, 2H), 2.69 (q, *J* = 7.2 Hz, 4H), 1.11 (t, *J* = 7.2 Hz,

6H).**^{13}C NMR** (126 MHz, $CDCl_3$) δ 149.45, 143.04, 128.57, 124.20, 121.83, 116.11, 113.35, 54.35, 47.70, 22.97, 11.34. **HRMS** (ESI+): calcd. for $C_{13}H_{18}N_3$ $[M+H]^+$ = 218.1657, found $[M+H]^+$ = 218.1664.

3-(2-(pyrrolidin-1-yl)ethyl)-1*H*-pyrrolo[2,3-b]pyridine (74) Synthesized according to general procedure C, Method I. Using 3-(2-bromoethyl)-1*H*-pyrrolo[2,3-b]pyridine (prepared according to the description in Scheme 9,[67,68] 30 mg, 0.13 mmol, 1 equiv.), 0.67 mL of MeOH (0.2 M) and pyrrolidine (1.33 mmol, 0.13 mL, 10 equiv.). The product was isolated on column chromatography (1:1 EtOAc:Hexanes and 2% TEA) to yield 19 mg of the product, 68%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.15 (dd, *J* = 4.8, 1.6 Hz, 1H), 8.00 (dd, *J* = 7.9, 1.5 Hz, 1H), 7.21 (d, *J* = 1.0 Hz, 1H), 7.07 (dd, *J* = 7.8, 4.8 Hz, 1H), 3.01 – 2.91 (m, 2H), 2.83 – 2.75 (m, 2H), 2.68 – 2.55 (m, 4H), 1.88 – 1.77 (m, 4H).**^{13}C NMR** (126 MHz, $CDCl_3$) δ 149.61, 143.20, 128.82, 124.36, 121.97, 116.27, 113.45, 58.28, 55.16, 25.72, 24.37. **HRMS** (ESI+): calcd. for $C_{13}H_{18}N_3$ $[M+H]^+$ = 216.1501, found $[M+H]^+$ = 216.1504.

3-(2-(piperidin-1-yl)ethyl)-1*H*-pyrrolo[2,3-b]pyridine (75) Synthesized according to general procedure C, Method I. Using 3-(2-bromoethyl)-1*H*-pyrrolo[2,3-b]pyridine (prepared according to the description in Scheme 9,[67,68] 30 mg, 0.13 mmol, 1 equiv.), 0.67 mL of MeOH (0.2 M) and piperidine (1.33 mmol, 0.13 mL, 10 equiv.). The product was isolated on column chromatography (1:1 EtOAc:Hexanes and 2% TEA) to yield 21 mg of the product, 71%. **^{1}H NMR** (400 MHz, MeOD) δ 8.15 (dd, *J* = 4.8, 1.5 Hz, 1H), 7.99 (dd, *J* = 7.9, 1.6 Hz, 1H), 7.19 (d, *J* = 1.0 Hz, 1H), 7.06 (dd, *J* = 7.8, 4.8 Hz, 1H), 2.99 – 2.85 (m, 2H), 2.67 – 2.59 (m, 2H), 2.58 – 2.46 (m, 4H), 1.69 – 1.58 (m, 4H), 1.52 – 1.42 (m, 2H). **^{13}C NMR** (126 MHz, MeOD) δ 148.05, 141.63, 127.25, 122.76, 120.43, 114.70, 111.95, 59.56, 54.00, 25.13, 23.80, 21.92. **HRMS** (ESI+): calcd. for $C_{14}H_{20}N_3$ $[M+H]^+$ = 230.1657, found $[M+H]^+$ = 230.1661.

2-(5-methoxybenzofuran-3-yl)-*N,N*-dimethylethan-1-amine (76) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxybenzofuran[65] (50 mg, 0.20 mmol, 1 equiv., prepared according to procedures A-B) and 2 M dimethylamine solution in MeOH (1 mL, 2.0 mmol, 10 equiv.). The product was isolated on preparative TLC (1:1 EtOAc:Hexanes) to yield 31 mg of the product, 73%. **^{1}H NMR** (500 MHz, MeOD) δ 7.51 (d, *J* = 1.1 Hz, 1H), 7.30 (d, *J* = 8.9 Hz, 1H), 7.05 (d, *J* = 2.6 Hz, 1H), 6.86 (dd, *J* = 8.9, 2.6 Hz, 1H), 3.81 (s, 3H), 2.85 – 2.78 (m, 2H), 2.67 – 2.60 (m, 2H), 2.32 (s, 6H). **^{13}C NMR** (126 MHz, MeOD) δ 157.33, 151.69, 143.75, 129.78, 119.28, 113.93, 112.67, 102.98, 59.76, 56.30, 45.31, 22.48. **HRMS** (ESI+): calcd. for $C_{13}H_{18}NO_2$ $[M+H]^+$ = 220.1338, found $[M+H]^+$ = 220.1348.

1-(2-(5-methoxybenzofuran-3-yl)ethyl)pyrrolidine (77) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxybenzofuran[65] (50 mg, 0.20 mmol, 1 equiv., prepared according to procedures A-B), 1 mL of MeOH (0.2 M), and pyrrolidine (0.19 mL, 2.0 mmol, 10 equiv.). The product was isolated on preparative TLC (1:1 EtOAc:Hexanes) to yield 33 mg of the product, 67%. **^{1}H NMR** (500 MHz, MeOD) δ 7.54 (d, *J* = 1.1 Hz, 1H), 7.32 (d, *J* = 8.9 Hz, 1H), 7.07 (d, *J* = 2.6 Hz, 1H), 6.88 (dd, *J* = 8.9, 2.6 Hz, 1H), 3.84 (s, 3H), 2.91 – 2.76 (m, 4H), 2.70 – 2.59 (m, 4H), 1.91 – 1.79 (m, 4H). **^{13}C NMR** (126 MHz, MeOD) δ 157.51, 151.87, 143.89, 129.97, 119.66, 114.10, 112.84, 103.19, 56.96, 56.48, 55.11, 24.37, 24.02. **HRMS** (ESI+): calcd. for $C_{15}H_{20}NO_2$ $[M+H]^+$ = 246.1494, found $[M+H]^+$ = 246.1502.

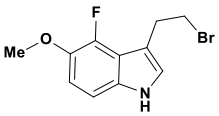
1-(2-(5-methoxybenzofuran-3-yl)ethyl)piperidine (78) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxybenzofuran[65] (50 mg, 0.20 mmol, 1 equiv., prepared according to procedures A-B), 1 mL of MeOH (0.2 M), and piperidine (0.19 mL, 2.0 mmol, 10 equiv.). The product was isolated on preparative TLC (1:1 EtOAc:Hexane) to yield 37 mg of the product, 73%. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.38 (s, 1H), 7.27 (d, *J* = 8.6 Hz, 1H), 7.02 (d, *J* = 2.6 Hz, 1H), 6.82 (dd, *J* = 8.9, 2.6 Hz, 1H), 3.80 (s, 3H), 2.91 (t, *J* = 8.2 Hz, 2H), 2.78 – 2.67 (m, 2H), 2.64 – 2.55 (m, 4H), 1.78 – 1.62 (m, 4H), 1.55 – 1.40 (m, 2H). **^{13}C NMR** (101 MHz, $CDCl_3$) δ 155.25, 149.60, 141.77, 127.88, 117.29, 112.44, 111.30, 101.56, 57.60, 55.49, 53.68, 24.61, 23.27, 20.28. **HRMS** (ESI+): calcd. for $C_{16}H_{22}NO_2$ $[M+H]^+$ = 260.1650, found $[M+H]^+$ = 260.1655.

1-(2-(5-methoxybenzofuran-3-yl)ethyl)-1,2,3,6-tetrahydropyridine (79) Synthesized according to general procedure C Method I. Using 3-(2-bromoethyl)-5-methoxybenzofuran[65] (500 mg, 1.97 mmol, 1 equiv., prepared according to procedures A-B), 2 mL of MeOH (1 M), and 1,2,3,6-tetrahydropyridine (0.36 mL, 3.9 mmol, 2 equiv.). The product was isolated on silica gel column chromatography (1:1 EtOAc:Hexane) to yield 479 mg of the product, 94%. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.45 (d, *J* = 1.1 Hz, 1H), 7.34 (d, *J* = 8.9 Hz, 1H), 7.02 (d, *J* = 2.6 Hz, 1H), 6.89 (dd, *J* = 8.9, 2.6 Hz, 1H), 5.79 (dtt, *J* = 9.4, 3.6, 2.1 Hz, 1H), 5.71 (dtt, *J* = 10.1, 3.3, 1.9 Hz, 1H), 3.85 (s, 3H), 3.12 – 3.06 (m, 2H), 2.93 – 2.86 (m, 2H), 2.79 – 2.72 (m, 2H), 2.67 (t, *J* = 5.7 Hz, 2H), 2.28 – 2.20 (m, 2H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 155.58, 150.06, 142.14, 128.57, 125.14, 124.96, 118.40, 112.62, 111.72, 102.02, 57.81, 55.83, 52.53, 49.96, 26.05, 21.60. **HRMS** (ESI+): calcd. for $C_{16}H_{20}NO_2$ $[M+H]^+$ = 258.1494, found $[M+H]^+$ = 258.1490.

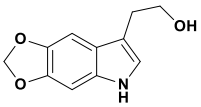
2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethan-1-ol (80) Synthesized according to general procedure A method II using 4-fluoro-5-methoxy-1*H*-indole. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.09 (s, 1H), 7.00 (d, *J* = 8.7 Hz, 1H), 6.91 (t, *J* = 8.2 Hz, 1H), 4.06 (t, *J* = 5.4 Hz, 2H), 3.95 (s, 3H), 3.00 (t, *J* = 5.5 Hz, 2H). **^{13}C NMR** (126 MHz,

$CDCl_3$) δ 147.78, 145.83, 139.52, 139.45, 134.61, 134.52, 124.47, 124.31, 117.21, 117.08, 112.16, 110.25, 110.23, 106.70, 106.67, 63.00, 59.05, 29.79.

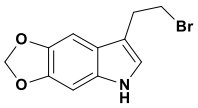
3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (81) Synthesized according to general procedure B using 2-(4-fluoro-5-methoxy-1*H*-indol-3-yl)ethan-1-ol. **^{1}H NMR** (500 MHz, $CDCl_3$) δ 8.40 (s, 1H), 7.05 – 6.98 (m, 2H), 6.94 (t, *J* = 8.3 Hz, 1H), 4.04 – 3.85 (m, 5H), 3.09 (t, *J* = 6.3 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 147.42, 145.47, 139.76, 139.71, 139.69, 133.06, 132.97, 116.88, 116.73, 111.47, 105.80, 105.77, 104.28, 58.83, 27.45, 23.34. **^{19}F NMR** (376 MHz, MeOD) δ -149.76.



2-(5*H*-[1,3]dioxolo[4,5-]indol-7-yl)ethan-1-ol (82) Synthesized according to general procedure A method I using 5*H*-[1,3]dioxolo[4,5-*f*]indole. **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.89 (br s, 1H), 7.00 – 6.91 (m, 2H), 6.82 (s, 1H), 5.93 (s, 2H), 3.60 (t, *J* = 7.6 Hz, 2H), 3.30 – 3.15 (m, 2H).

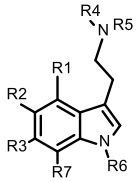


3-(2-bromoethyl)-4-fluoro-5-methoxy-1*H*-indole (83) Synthesized according to general procedure B using 5*H*-[1,3]dioxolo[4,5-f]indole. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.89 (s, 1H), 6.99 – 6.90 (m, 2H), 6.82 (s, 1H), 5.94 (s, 2H), 3.60 (t, *J* = 7.3 Hz, 2H), 3.25 (t, *J* = 0.8 Hz, 2H).**^{13}C NMR** (126 MHz, $CDCl_3$) δ 144.60, 142.44, 131.46, 121.08, 120.86, 112.50, 100.17, 96.21, 91.63, 32.33, 29.27.



Supplementary Table 1: EC50 and Efficacy Summary of Tryptamine Analogs and Commercial Ligands

Tryptamine Analogs

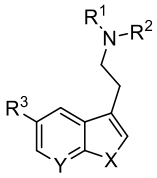
Structure									Activity					
#	Name 	R^1	R^2	R^3	R^4	R^5	R^6	R^7	5-HT1A cAMP	5-HT1A BRET Gi1	5-HT2A BRET Gq	5-HT1A cAMP	5-HT1A BRET Gi1	5-HT2A BRET Gq
									EC50 in nM (pEC50, SEM)			Efficacy as %5HT (SEM)		
1	**DMT**	H	H	H	Me	Me	H	H	651.63 (6.19, 0.14)	1584.89 (5.80, 0.02)	94.41 (7.03,0.16)	98.3 (5.94)	95.1 (1.01)	67.2 (2.51)
2	**MET**	H	H	H	Me	Et	H	H	302.00 (6.52, 0.08)	-	-	90.0 (4.09)	-	-
3	**MiPT**	H	H	H	Me	iPr	H	H	1976.97 (5.70, 0.04)	-	-	73.5 (13.60)	-	-
4	**DiPT**	H	H	H	iPr	iPr	H	H	1327.39 (5.88, 0.05)	-	-	59.8 (1.14)	-	-
5	**PyrT**	H	H	H	Pyrrolidine		H	H	19.10 (7.72, 0.20)	-	-	99.0 (4.33)	-	-
6	**PipT**	H	H	H	Piperidine		H	H	485.29 (6.31, 0.27)	-	-	88.2 (18.97)	-	-
7	**desethylisoquinuclidineT**	H	H	H	desethyl-isoquinuclidine		H	H	15454.5 (4.81, 0.18)	-	-	75.0 (21.47)	-	-
8	**5-MeO-DMT**	H	OMe	H	Me	Me	H	H	3.67 (8.44, 0.11)	25.59 (7.59,0.06)	24.83 (7.61, 0.12)	104.2 (4.78)	105.2 (1.52)	85.5 (0.93)
9	**5-MeO-MET**	H	OMe	H	Me	Et	H	H	4.44 (8.35, 0.06)	25.94 (7.59, 0.06)	37.67 (7.42, 0.06)	95.6 (1.93)	104.1 (3.95)	74.8 (3.24)
10	**5-MeO-MPT**	H	Ome	H	Me	Pr	H	H	18.58 (7.73, 0.14)	60.26 (7.22, 0.07)	28.31 (7.55, 0.04)	105.5 (4.16)	96.4 (10.97)	80.1 (2.60)
11	**5-MeO-DET**	H	Ome	H	Et	Et	H	H	11.04 (7.96, 0.15)	37.07 (7.43, 0.01)	171.79(6.77, 0.18)	118.8 (2.44)	101.9 (2.65)	75.0 (6.13)
12	**5-MeO-EPT**	H	Ome	H	Et	Pr	H	H	2.47 (8.61, 0.03)	8.36 (8.08, 0.03)	22.08 (7.66, 0.06)	105.2 (4.09)	100.4 (5.01)	81.8 (5.70)
13	**5-MeO-DPT**	H	Ome	H	Pr	Pr	H	H	3.87 (8.41, 0.08)	15.42 (7.81, 0.01)	20.18 (7.70, 0.06)	110.7 (3.17)	101.5 (2.28)	84.9 (3.63)
14	**5-MeO-MiPT**	H	OMe	H	Me	iPr	H	H	51.76 (7.29, 0.04)	112.46 (6.95, 0.04)	17.34 (7.76, 0.01)	100.1 (2.11)	94.8 (7.94)	88.3 (3.07)
15	**5-MeO-PriPT**	H	OMe	H	Pr	iPr	H	H	26.67 (7.57, 0.12)	101.16 (7.00, 0.01)	51.17 (7.29, 0.02)	105.4 (5.96)	100.5 (0.87)	88.9 (2.67)
16	**5-MeO-MALT**	H	OMe	H	Me	Allyl	H	H	9.64 (8.02, 0.20)	42.66 (7.37,0.07)	24.47 (7.59, 0.14)	107.2 (5.91)	99.7 (6.52)	85.3 (0.63)
17	**5-MeO-DALT**	H	OMe	H	Allyl	Allyl	H	H	4.66 (8.33, 0.06)	-	-	94.6 (3.64)	-	-
18	**5-MeO-PyrT**	H	OMe	H	Pyrrolidine		H	H	0.16 (9.78, 0.09)	2.13 (8.67,0.05)	81.47 (7.09, 0.10)	102.8 (3.13)	103.7 (1.51)	87.9 (1.75)
19	**5-MeO-α-Me-PyrT**	H	OMe	H	α-Me-Pyrrolidine		H	H	2.40 (8.62, 0.24)	-	-	108.3 (5.34)	-	-
20	**5-MeO-PipT**	H	OMe	H	Piperidine		H	H	31.55 (7.50, 0.16)	88.51 (7.05, 0.01)	ND*	106.3 (7.13)	103.8 (2.07)	ND*

21	**5-MeO-3-PyrrolineT**	H	OMe	H	3-Pyrroline		H	H	0.35 (10.59, 0.20)	0.26 (9.59, 0.02)	9.91 (8.00, 0.12)	101.3 (2.70)	101.3 (2.18)	95.0, 3.50
22	**5-MeO-ImidT**	H	OMe	H	Imidazole		H	H	292.42(6.53, 0.17)	-	-	69.6 (9.33)	-	-
23	**5-MeO-dehydro-PipT**	H	OMe	H	1,2,3,6-tetrahydropyridine		H	H	8.77 (8.06, 0.14)	-	-	105.1 (2.15)	-	-
24	**5-MeO-desethylisoquinuclidineT**	H	OMe	H	desethyl-isoquinuclidine		H	H	628.06(6.20, 0.12)	-	-	93.6 (3.98)	-	-
25	**4-F, 5-MeO-DMT**	F	OMe	H	Me	Me	H	H	0.19 (9.73, 0.12)	1.77 (8.75, 0.04)	86.50 (7.06, 0.08)	110.2 (2.08)	103.5 1.59)	68.8 (2.08)
26	**4-F, 5-MeO-MET**	F	OMe	H	Me	Et	H	H	0.32 (9.50, 0.39)	1.77 (8.75, 0.10)	99.31 (7.00, 0.21)	108.5 (1.50)	101.5 (0.72)	70.0 (4.27)
27	**4-F, 5-MeO-DET**	F	OMe	H	Et	Et	H	H	0.21 (9.69, 0.10)	3.18 (8.50,0.04)	672.98 (6.17, 0.18)	110.5 (1.70)	104.0 (1.68)	69.0 (12.80)
28	**4-F, 5-MeO-MPT**	F	OMe	H	Me	Pr	H	H	0.42 (9.38, 0.27)	3.06 (8.51, 0.03)	114.55 (6.94, 0.01)	112.5 (2.80)	103.9 (0.92)	79.0 (0.60)
29	**4-F, 5-MeO-EPT**	F	OMe	H	Et	Pr	H	H	0.15 (9.82, 0.28)	0.95 (9.02, 0.10)	154.53 (6.81, 0.03)	112.9 (3.72)	99.7 (4.61)	90.1 (1.42)
30	**4-F, 5-MeO-DPT**	F	OMe	H	Pr	Pr	H	H	1.31 (8.88, 0.24)	3.70 (8.43,0.11)	142.23 (6.85, 0.05)	106.2 (0.99)	98.2 (1.89)	95.3 (1.09)
31	**4-F, 5-MeO-MiPT**	F	OMe	H	Me	iPr	H	H	2.98 (8.53, 0.14)	14.16 (7.85, 0.08)	151.71 (6.82, 0.003)	109.0 (3.99)	103.5 0.47)	83.4 (7.03)
32	**4-F, 5-MeO-PriPT**	F	OMe	H	Pr	iPr	H	H	5.96 (8.23, 0.12)	16.79 (7.78, 0.09)	249.46 (6.60,0.31)	104.4 (0.49)	100.6 (0.72)	86.3 (5.33)
33	**4-F, 5-MeO-DiPT**	F	OMe	H	iPr	iPr	H	H	6.00 (8.22, 0.07)	35.81 (7.45, 0.004)	338.06 (6.47, 0.19)	98.5 (1.30)	100.1 (1.89)	96.9 (5.69)
34	**4-F, 5-MeO-MALT**	F	OMe	H	Me	Allyl	H	H	2.57 (8.59, 0.22)	8.41 (8.08, 0.05)	156.31 (6.81, 0.27)	104.7 (1.90)	100.0 (2.84)	66.8 (9.87)
35	**4-F, 5-MeO-PyrT**	F	OMe	H	Pyrrolidine		H	H	0.03 (10.55, 0.08)	0.37 (9.43, 0.07)	299.92 (6.52, 0.04)	105.9 (1.07)	102.8 (1.62)	83.6 (2.45)
36	**4-F, 5-MeO-PipT**	F	OMe	H	Piperidine		H	H	2.42 (8.62. 0.19)	8.97 (8.05, 0.003)	ND*	110.8 (2.41)	105.3 (1.46)	ND*
37	**4-F, 5-MeO-3-PyrrolineT**	F	OMe	H	3-Pyrroline		H	H	0.0023 (11.64, 0.70)	0.22 (9.66, 0.32)	120.22 (6.92, 0.10)	111.1 (2.94)	105.9 (0.62)	95.6 (3.72)
38	**4-F, 5-MeO-desethyl isoquinuclidine**	F	OMe	H	desethyl-isoquinuclidine		H	H	173.38 (6.76, 0.29)	-	-	106.8 (10.56)	-	-
39	**4-F, 5-MeO-ethyl isoquinuclidine**	F	OMe	H	ethyl-isoquinuclidine		H	H	47.32 (7.33, 0.11)	-	-	96.8 (2.29)	-	-
40	**4-F, 5-OH-DMT**	F	OH	H	Me	Me	H	H	0.25 (9.60, 0.82)	1.13 (8.95, 0.10)	7.46 (8.13, 0.08)	105.1 (2.11)	101.0 (2.74)	71.0 (4.01)
41	**4-F, 5-OH-DET**	F	OH	H	Et	Et	H	H	0.76 (9.12, 0.14)	-	-	106.4 (3.58)	-	
42	**4-F, 5-OH-PyrT**	F	OH	H	Pyrrolidine		H	H	0.040 (10.38,0.11)	-	-	110.3 (2.53)	-	-
43	**1-Me-indole-5-MeO-PyrT**	H	OMe	H	Pyrrolidine		Me	H	89.13 (7.05, 0.20)	-	-	104.9 (8.52)	-	-
44	**1-Me-indole-5-MeO-3-PyrrolineT**	H	OMe	H	3-Pyrroline		Me	H	209.41 (6.68, 0.21)	-	-	99.4 (4.26)	-	-
45	**1-Me-indole-5-MeO-PyrroleT**	H	OMe	H	Pyrrole		Me	H	6576.58 (5.18, 0.0004)	-	-	96.4 (5.25)	-	-
46	**1-Me-indole-4-F, 5-MeO-PyrT**	F	OMe	H	Pyrrolidine		Me	H	100.00(7.00, 0.10)	-	-	110.9 (6.23)	-	-

47	**1-Me-indole-4-F, 5-MeO-3-PyrrolineT**	F	OMe	H	3-Pyrroline		Me	H	2.74 (8.56, 0.15)	-	-	107.5 (0.92)	-	-
48	**5-Bnz-PyrT**	H	OBz	H	Pyrrolidine		H	H	33.04 (7.48, 0.05)	-	-	109.6 (4.47)	-	-
49	**Bufotenine (5-OH-DMT)**	H	OH	H	Me	Me	H	H	-	9.29 (8.03, 0.01)	27.86 (7.56, 0.04)	-	104.5 (0.65)	74.6 (4.30)
50	**5-OH-PyrT**	H	OH	H	Pyrrolidine		H	H	0.10 (10.01, 0.34)	-	-	102.8 (2.16)	-	-
51	**5-OH-desethylisoquinuclidineT**	H	OH	H	desethylisoquinuclidine		H	H	149.28 (6.83, 0.20)	-	-	106.6 (5.46)	-	-
52	**Psilocin**	OH	H	H	Me	Me	H	H	511.68 (6.29, 0.11)	1083.93 (5.97, 0.13)	35.89 (7.45, 0.05)	102.0 (4.48)	88.7 (4.78)	52.6 (7.70)
53	**4-OH-MET**	OH	H	H	Me	Et	H	H	769.19 (6.11, 0.06)	-	-	85.1 (6.93)	-	-
54	**4-MeO-DMT**	Ome	H	H	Me	Me	H	H	558.47 (6.25, 0.23)	-	-	100.2 (11.45)	-	-
55	**4-MeO-PyrT**	Ome	H	H	Pyrrolidine		H	H	163.68 (6.79, 0.30)	-	-	107.0 (3.58)	-	-
56	**4-MeO-3-PyrrolineT**	OMe	H	H	3-Pyrroline		H	H	0.22 (8.47, 0.16)	-	-	106.2 (2.17)	-	-
57	**4-F-DMT**	F	H	H	Me	Me	H	H	87.70 (7.06, 0.18)	-	-	101.8 (2.94)	-	-
58	**4-F-PyrT**	F	H	H	Pyrrolidine		H	H	16.75 (7.78, 0.08)	-	-	109.1 (3.46)	-	-
59	**4-F,3-PyrrolineT**	F	H	H	3-Pyrroline		H	H	0.88 (9.06, 0.06)		-	106.7 (2.34)	-	-
60	**6-MeO-DMT**	H	H	OMe	Me	Me	H	H	3235.94 (5.49, 0.11)	-	-	99.7 (2.66)	-	-
61	**6-MeO-PyrT**	H	H	OMe	Pyrrolidine		H	H	497.73 (6.30, 0.05)	-	-	111.0 (7.06)	-	-
62	**6-MeO-3-PyrrolineT**	H	H	OMe	3-Pyroline		H	H	37.50 (7.43, 0.09)	-	-	100.5 (3.95)	-	-
63	**5,6-methylenedioxy-DMT**	H	-O-CH_2-O-		Me	Me	H	H	64.42 (7.19. 0.01)	-	-	105.0 (9.93)	-	-
64	**5,6-methylenedioxy-PyrT**	H	-O-CH_2-O-		Pyrrolidine		H	H	3.50 (8.46, 0.09)	-	-	105.3 (5.78)	-	-
65	**5,6-methylenedioxy-desethyl-isoquinuclidine**	H	-O-CH_2-O-		desethyl-isoquinuclidine		H	H	9885.53 (5.01, 0.07)	-	-	115.8 (2.48)	-	-
66	**5,6-methylenedioxy-isoquinuclidine**	H	-O-CH_2-O-		ethyl-isoquinuclidine		H	H	1690.44(5.77, 0.25)	-	-	88.7 (23.74)	-	-
67	**5-MeS-DMT**	H	5-SMe	H	Me	Me	H	H	2.01 (8.70, 0.08)	11.69 (7.93, 0.07)	30.06 (7.52, 0.08)	102.0 (2.63)	106.6 (0.99)	73.1 (0.87)
68	**5-MeS-MET**	H	5-SMe	H	Me	Et	H	H	3.35 (8.48, 0.08)	-	-	101.9 (1.93)	-	-
69	**5-MeS-PyrT**	H	5-SMe	H	Pyrrolidine		H	H	0.51 (9.30, 0.12)	1.05 (8.98, 0.05)	174.58 (6.76, 0.03)	103.7 (0.64)	99.5 (2.03)	78.9 (2.97)
70	**7-Me-indole-DMT**	H	H	H	Me	Me	H	Me	1207.81(5.92, 0.40)	-	-	91.5 (15.22)	-	-
71	**5-HT**	H	OH	H	H	H	H	N	0.21 (9.68, 0.04)	2.61 (8.58, 0.04)	4.40 (8.36, 0.07)	100.0	100.0	100.0

* not detected – out of assay range

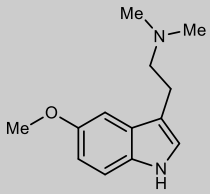
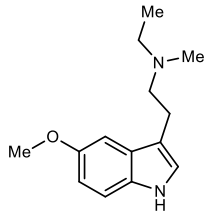
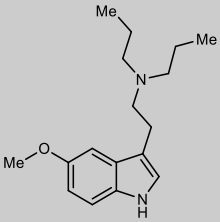
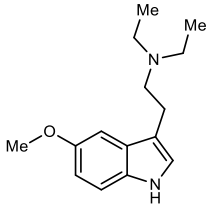
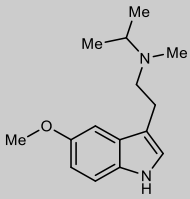
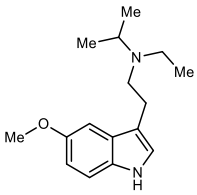
Azaindoles and Benzofurans

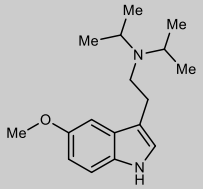
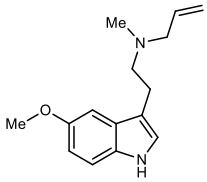
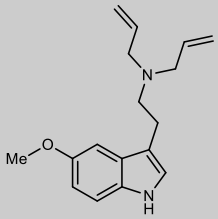
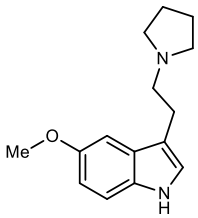
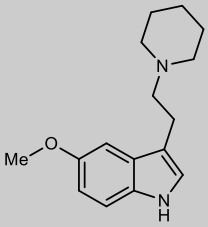
Structure							Activity	
#	Name	R_1	R_2	R_3	X	Y	5-HT1A cAMP	
							EC50 in nM (pEC50, SEM)	Efficacy as %5HT (SEM)
72	**7-azaindole-DMT**	Me	Me	H	N	N	1905.46 (5.72, 0.25)	88.3 (20.41)
73	**7-azaindole-DET**	Et	Et	H	N	N	7798.30 (5.11, 0.22)	108.8 (5.02)
74	**7-azaindole-PyrT**	Pyrrolidine		H	N	N	3243.40 (5.49, 0.14)	95.6 (2.11)
75	**7-azaindole-PipT**	Piperidine		H	N	N	18492.69 (4.73, 0.01)	101.5 (0.02)
76	**5-MeO-benzofuran-DMT**	Me	Me	OMe	O	C	178.24 (6.75, 0.18)	105.4 (1.13)
77	**5-MeO-benzofuran-PyrT**	Pyrrolidine		Ome	O	C	7.19 (8.14, 0.23)	112.8 (2.95)
78	**5-MeO-benzofuran-PipT**	Piperidine		Ome	O	C	415.91 (6.38, 0.26)	114.1 (8.86)
79	**5-MeO-benzofuran-dehydro-PipT**	1,2,3,6-tetrahydropyridine		Ome	O	C	7128.53 (5.15, 0.18)	125.9 (9.71)
71	**5-HT**	H	H	OH	N	C	0.21 (9.68, 0.04)	100.0

Commercial Ligands

Activity							
#	Name	5-HT1A cAMP	5-HT1A BRET Gi1	5-HT2A BRET Gq	5-HT1A cAMP	5-HT1A BRET Gi1	5-HT2A BRET Gq
		EC50 in nM (pEC50, SEM)			Efficacy as %5HT (SEM)		
80	**LSD**	0.26 (9.08, 0.26)	1.20 (8.92, 0.03)	0.25 (9.60, 0.19)	119.8 (7.02)	100.9 (1.73)	82.4 (3.23)
81	**Mescaline**	-	19588.45 (4.71, 0.33)	2404.36 (5.62, 0.15)	-	42.0 (6.38)	92.6 (1.36)
82	**8-OH-DPAT**	0.46 (10.41,0.46)	-	-	108.5 (12.09)	-	-
83	**SEP-856**	1081.43(5.97, 0.32)	-	-	70.4 (6.33)	-	-
84	**N-(1-Naphthyl)piperazine**	50.35 (7.30, 0.29)	-	-	137.0 (14.09)	-	-
85	**Vilazodone**	0.03 (10.60, 0.20)	0.48 (9.32, 0.57)		110.5 (3.72)	97.4 (0.46)	-
86	**Tandospirone**	1.51 (8.82, 0.21)	-	-	96.3 (4.50)	-	-
87	**Gepirone**	16.18 (7.79, 0.10)	-	-	95.4 (4.29)	-	-
88	**Buspirone**	1.44 (8.84, 0.10)	9.68 (8.01, 0.16)	-	96.0 (0.62)	93.4 (0.66)	-
89	**Aripiprazole**	-	5.83 (8.24, 0.002)	-	-	77.1 (2.72)	-
71	**5-HT**	0.21 (9.68, 0.04)	2.61 (8.58, 0.04)	4.40 (8.36, 0.07)	100.0	100.0	100.0

Supplementary Table 2 | Anectdotal Reports of "Designer" Tryptamine-Mediated Effects in Humans

Name	*Structure*	*References*
5-MeO-DMT		***https://erowid.org/experiences/subs/exp_5MeODMT.shtml*** ***https://isomerdesign.com/PiHKAL/read.php?domain=tk&id=38***
5-MeO-MET		***https://erowid.org/experiences/exp.cgi?S1=910&S2=-1&C1=-1&Str*** ***https://drugs-forum.com/threads/5-meo-met.40412/***
5-MeO-DPT		***https://erowid.org/experiences/exp.cgi?S1=391&OldSort=RA_PDD&NewSort=SA&Start=0&ShowViews=0&Cellar=0*** ***https://drugs-forum.com/threads/5-meo-dpt-trip-reports.26997/*** ***https://isomerdesign.com/PiHKAL/read.php?domain=tk&id=36***
5-MeO-DET		***https://erowid.org/experiences/exp.php?ID=43242*** ***https://erowid.org/library/books_online/tihkal/tihkal36.shtml***
5-MeO-MiPT		***https://erowid.org/experiences/subs/exp_5MeOMIPT.shtml*** ***https://isomerdesign.com/PiHKAL/read.php?domain=tk&id=40***
5-MeO-EiPT		***https://erowid.org/experiences/exp.cgi?S1=727&S2=-1&C1=-1&Str***

5-MeO-DiPT		***https://erowid.org/experiences/subs/exp_5MeODiPT.shtml*** ***http://isomerdesign.com/PiHKAL/read.php?id=37&domain=tk***
5-MeO-MALT		***https://erowid.org/experiences/exp.cgi?S1=883&S2=-1&C1=-1&Str***
5-MeO-DALT		***https://erowid.org/experiences/subs/exp_5MeODALT.shtml*** ***http://isomerdesign.com/PiHKAL/read.php?domain=tk&code=5-MEO-DALt***
5-MeO-PyrT		***https://isomerdesign.com/PiHKAL/read.php?domain=tk&id=43*** ***https://www.youtube.com/watch?v=x9WmEVRxebE&ab_channel=HamiltonMorris***
5-MeO-PipT		***https://erowid.org/experiences/exp.php?ID=115247*** ***https://www.reddit.com/r/researchchemicals/comments/ltyf38/5meopipt_trial_run_and_dosage_estimation/*** ***https://www.reddit.com/r/Psychedelics/comments/s8rg40/oral_low_to_moderate_dose_5meopipt_compared_with/***

Supplementary Table 3: Compound Potency and Efficacy at Mutant 5-HT1A and 5-HT2A Receptors

	5-HT1A WT		5-HT1A A365N$^{6.55}$		5-HT1A N386V$^{7.39}$	
	Gi1 BRET					
	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)
5-HT	2.61 (8.58, 0.04)	100.0	13.46 (7.87, 0.1)	100.0	26.24 (7.58, 0.04)	100.0
DMT	1584.89 (5.8, 0.02)	95.1 (1.01)	5382.7 (5.27, 0.09)	86.2 (3.62)	-	-
5-MeO-DMT	25.59 (7.59, 0.06)	105.2 (1.52)	196.79 (6.71, 0.11)	100 (1.78)	38.19 (7.42, 0.06)	99.4 (1.33)
4-F, 5-OH-DMT	1.13 (8.95, 0.1)	101 (2.74)	5.68 (8.25, 0.14)	102.7 (0.98)	-	-
4-F, 5-MeO-DMT	1.77 (8.75, 0.04)	103.5 (1.59)	20.14 (7.7, 0.14)	103.7 (1.16)	3.71 (8.43, 0.09)	102.4 (0.9)
5-MeO-PyrT	2.13 (8.67, 0.05)	103.7 (1.51)	28.91 (7.54, 0.09)	104.8 (0.85)	24.89 (7.6, 0.05)	101.7 (0.47)
4-F, 5-MeO-PyrT	0.37 (9.43, 0.07)	102.8 (1.62)	5.08 (8.29, 0.05)	101.9 (0.02)	3 (8.52, 0.07)	101.2 (2.08)
LSD	1.2 (8.92, 0.03)	100.9 (1.73)	1.73 (8.76, 0)	101.1 (1.2)	0.37 (9.44, 0.04)	92.3 (0.68)
Bufotenine (5-OH-DMT)	9.29 (8.03, 0.01)	104.5 (0.65)	171 (6.77, 0.01)	103.8 (1.38)	-	-
5-MeS-DMT	11.69 (7.93, 0.07)	106.6 (0.99)	30.55 (7.52, 0.06)	105.1 (1.19)	-	-
5-MeS-PyrT	1.05 (8.98, 0.05)	99.5 (2.03)	5.08 (8.29, 0.03)	101.1 (1.02)	-	-
	5-HT2A WT		**5-HT2A N343A$^{6.55}$**		**5-HT2A V366N$^{7.39}$**	
	Gq BRET					
	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)	EC50 in nM (pEC50, SEM)	Efficacy as %5-HT (SEM)
5-HT	4.4 (8.36, 0.07)	100.0	583.45 (6.23, 0.09)	100.0	15.38 (7.81, 0.14)	100.0
DMT	94.41 (7.03, 0.16)	67.2 (2.51)	479.73 (6.32, 0.13)	34.8 (11.09)	-	-
5-MeO-DMT	24.83 (7.61, 0.12)	85.5 (0.93)	68.55 (7.16, 0.08)	55.6 (6.58)	77.27 (7.11, 0.4)	126.6 (1.94)
4-F, 5-OH-DMT	7.46 (8.13, 0.08)	71 (4.01)	140.93 (6.85, 0.07)	57 (6.08)	-	-
4-F, 5-MeO-DMT	86.5 (7.06, 0.08)	68.8 (2.08)	140.6 (6.85, 0.25)	67.3 (6.04)	391.74 (6.41, 0.11)	101.6 (2.43)
5-MeO-PyrT	81.47 (7.09, 0.1)	87.9 (1.75)	111.94 (6.95, 0.1)	68.7 (5.2)	55.59 (7.26, 0.28)	150.3 (0.52)
4-F, 5-MeO-PyrT	299.92 (6.52, 0.04)	83.6 (2.45)	257.63 (6.59, 0.22)	77.8 (16.65)	157.04 (6.8, 0.15)	129.4 (5.89)
LSD	0.25 (9.6, 0.19)	82.4 (3.23)	0.07 (10.17, 0.1)	77.1 (6.05)	0.06 (10.2, 0.06)	111.4 (0.74)

Bufotenine (5-OH-DMT)	27.86 (7.56, 0.04)	74.6 (4.3)	279.25 (6.55, 0.36)	56.3 (3.36)	-	-
5-MeS-DMT	30.06 (7.52, 0.08)	73.1 (0.87)	26.49 (7.58, 0.18)	51.2 (1.63)	-	-
5-MeS-PyrT	174.58 (6.76, 0.03)	78.9 (2.97)	111.17 (6.95, 0.09)	64 (5.7)	-	-

	5-HT1A WT		**5-HT1A F361L$^{6.51}$**		
	cAMP				
	EC50 in nM (pEC50, SEM)	**Efficacy as %5-HT (SEM)**	**EC50 in nM (pEC50, SEM)**	**Efficacy as %5-HT (SEM)**	**Fold-Change (F361L EC50/WT EC50)**
5-HT	0.31 (-9.51, 0.09)	100.0	42.36 (-7.37, 0.06)	100.0	138
5-MeO-DMT	3.49 (-8.46, 0.19)	107.5 (6.04)	181.97 (-6.74, 0.02)	92.4 (2.71)	52
4-F, 5-MeO-PyrT	0.03 (-10.54, 0.07)	99.5 (0.99)	0.84 (-9.08, 0.04)	103 (1)	29

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